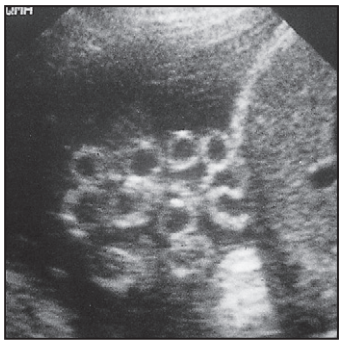
Example of antenatal diagnosis-gastroschisis



(a)



(b)

Figure 9.2 Gastroschisis on antenatal ultrasound showing free loops of small bowel in the amniotic fluid **(a)** and following delivery **(b)**. Antenatal diagnosis allowed the baby to be delivered at a paediatric surgical unit and the parents to be forewarned about the need for surgery. Satisfactory surgical repair was achieved. (Courtesy of Mr Karl Murphy.)

Case History

9.1 Antenatal diagnosis

A routine ultrasound scan at 18 weeks' gestation identified an abnormal 'lemon-shaped' skull (Fig. 9.3). This, together with an abnormal appearance of the cerebellum, is the Arnold–Chiari malformation, which is associated with spina bifida. An extensive spinal defect was confirmed on ultrasound. Dilatation of the cerebral ventricles and talipes already present in this fetus suggested a severe spinal lesion. After counselling, the parents decided to terminate the pregnancy.



Figure 9.3 Transverse section showing a 'lemon-shaped' skull on ultrasound instead of the normal oval shape. This is associated with spina bifida. (Courtesy of Mr Guy Thorpe-Beeston.)

the incidence of respiratory distress syndrome (RDS) (relative risk 0.66), of intraventricular haemorrhage (relative risk 0.54) and neonatal mortality (relative risk 0.69) in preterm infants. For optimal effect, a completed course needs to be given at least 24 h before delivery.
- *Digoxin or flecainide* can be given to the mother to treat fetal supraventricular tachycardia.

There are a few conditions where therapy can be given to the fetus directly:

- *Rhesus isoimmunisation*. Severely affected fetuses become anaemic and may develop *hydrops fetalis*, with oedema and ascites. Infants at risk are identified by maternal antibody screening. Regular ultrasound of the fetus is performed to detect fetal anaemia non-invasively using Doppler velocimetry of the fetal middle cerebral artery. Fetal blood transfusion via the umbilical vein may be required regularly from about 20 weeks' gestation. The incidence of rhesus haemolytic disease has fallen markedly since anti-D immunisation of mothers was introduced but hydrops fetalis is still seen due to other red blood cell antibodies such as Kell.
- *Perinatal isoimmune thrombocytopenia*. This condition is analogous to rhesus isoimmunisation but involves maternal antiplatelet antibodies crossing the placenta. It is rare, affecting about 1 in 5000 births. Intracranial haemorrhage secondary to fetal thrombocytopenia occurs in up to 25%. The problem may be anticipated if there was a previously affected infant, when prenatal intravenous immunoglobulin can be given or repeated intrauterine platelet transfusions performed.

Maternal glucocorticoid therapy before preterm delivery markedly reduces morbidity and mortality in the neonate.

Fetal surgery

Fetal surgery is a relatively new development with varying results. Procedures which have been performed include:

- Catheter shunts inserted under ultrasound guidance. This is to drain fetal pleural effusions (pleuro-amniotic shunts), often from a chylothorax (lymphatic fluid) or congenital cystic adenomatous malformation of the lung. One end of a looped catheter lies in the chest, the other end in the amniotic cavity.
- Laser therapy to ablate placental anastomoses which lead to the twin–twin transfusion syndrome (TTTS)

- Intrauterine shunting for obstruction to urinary outflow as with posterior urethral valves
- Dilatation of stenotic heart valves via a transabdominal catheter inserted under ultrasound guidance into the fetal heart. Results appear promising
- Endotracheal balloon occlusion for congenital diaphragmatic hernia, as tracheal obstruction in utero may promote lung growth
- Surgical correction by hysterotomy. This is when the uterus is opened at 22–24 weeks' gestation. It has been performed in a few specialist centres for spina bifida but may precipitate preterm delivery and its efficacy remains highly uncertain. Results of fetal surgery to close spina bifida suggest that hydrocephalus may be reduced but does not improve the prognosis of the spinal lesion.

Outcome has mostly been very poor because of the severity of the conditions treated. Careful case selection and follow-up are required to ensure that these novel forms of treatment are of long-term benefit.

Obstetric conditions affecting the fetus

Pre-eclampsia

Mothers with pre-eclampsia may require preterm delivery because of the maternal risks of eclampsia and of cerebrovascular accident or the fetal risks associated with placental insufficiency and growth restriction. Determining the optimal time for preterm delivery requires an evaluation of the risk to the mother and fetus of allowing the pregnancy to continue compared with the neonatal complications associated with preterm birth.

Placental insufficiency and intrauterine growth restriction (IUGR)

Fetal growth may be progressively restricted because of placental insufficiency. Transfer of oxygen and nutrients is reduced. The growth-restricted fetus will need to be monitored closely to prevent intrauterine death. This is done by measuring growth parameters, the biophysical profile (amniotic fluid volume, fetal movement, fetal tone, fetal breathing movements, fetal heart activity) and Doppler blood flow velocity (umbilical and middle cerebral artery). Absence or reversal of flow velocity during diastole carries an increased risk of morbidity from hypoxic damage to the gut or brain, or of intrauterine death. These measurements assist in deciding the optimal time for delivery of a growth-restricted fetus.

Multiple births

Twins occur naturally in the UK in 1 in 90 deliveries, triplets in 1 in 90^2, i.e. approximately 1 in 8000, and quadruplets in 1 in 90^3, i.e. approximately 1 in every 700 000 deliveries. Over the last decade, the number of triplets and higher-order births has more than doubled, mainly from assisted reproduction programmes and advancing maternal age. One in 70 births is now a multiple birth, although the number of triplets and higher-order births has recently declined in the UK.

The main problems for the infant associated with multiple births are:

- *Preterm labour*. The median gestation for twins is 37 weeks, for triplets 34 weeks and for quads 32 weeks. Preterm delivery is the most important cause of the greater perinatal mortality of multiple births, especially for triplets and higher-order pregnancies. When a higher-order pregnancy is identified, embryo reduction may be offered.
- *Intrauterine growth restriction (IUGR)*. Fetal growth in one or more fetuses may deteriorate and needs to be monitored regularly.
- *Congenital abnormalities*. These occur twice as frequently as in a singleton, but the risk is increased four-fold in monochorionic twins.
- *Twin–twin transfusion syndrome (TTTS) in monochorionic twins (shared placenta).* May cause extreme preterm delivery, fetal death and discrepancy in growth.
- *Complicated deliveries*, e.g. due to malpresentation of the second twin at vaginal delivery.

Finding sufficient intensive care cots for preterm multiple births can be problematic.

Although multiple births may look endearing, the families may need additional assistance and support:

- Practical – with their care and housework (requires about 200 h/week for triplets in infancy!)
- Emotional and physical exhaustion
- Loss of privacy as a couple
- Additional financial costs
- Increased behavioural problems in the infants and their siblings. While being a multiple birth may provide companionship, affection and stimulation between each other, it may also engender domination, dependency and jealousy.

There are local and national support groups for parents of multiple births.

Summary

Multiple births

- have markedly increased over the last 20 years
- are associated with an increased risk of prematurity, intrauterine growth restriction (IUGR), congenital malformations and twin–twin transfusion syndrome (in monochorionic twins)
- are responsible for 30% of very low birthweight infants (<1.5 kg birthweight)
- provide many additional problems for their parents to care for them.

Maternal conditions affecting the fetus

Diabetes mellitus

Women with insulin-dependent diabetes find it more difficult to maintain good diabetic control during pregnancy and have an increased insulin requirement. Poorly-controlled maternal diabetes is associated with polyhydramnios and pre-eclampsia, increased rate of early fetal loss, congenital malformations and late unexplained intrauterine death. Ketoacidosis carries a high fetal mortality. With meticulous attention to diabetic control, the perinatal mortality rate is now only slightly greater than in non-diabetics. The National Institute for Health and Clinical Excellence (NICE) has produced guidance on the management of diabetes and its complications from preconception to the postnatal period. The emphasis is on aiming for good control of blood glucose.

Fetal problems associated with maternal diabetes are:

- *Congenital malformations*. Overall, there is a 6% risk of congenital malformations, a three-fold increase compared with the non-diabetic population. The range of anomalies is similar to that for the general population, apart from an increased incidence of cardiac malformations, sacral agenesis (caudal regression syndrome) and hypoplastic left colon, although the latter two conditions are rare. Studies show that good diabetic control periconceptionally reduces the risk of congenital malformations.
- *Intrauterine growth restriction (IUGR)*. There is a three-fold increase in growth restriction in mothers with long-standing microvascular disease.
- *Macrosomia* (Fig. 9.4). Maternal hyperglycaemia causes fetal hyperglycaemia as glucose crosses the placenta. As insulin does not cross the placenta, the fetus responds with increased secretion of insulin, which promotes growth by increasing both cell number and size. About 25% of such infants have a birthweight greater than 4 kg compared with 8% of non-diabetics. The macrosomia predisposes to cephalopelvic disproportion, birth asphyxia, shoulder dystocia and brachial plexus injury.



Figure 9.4 Infant of a diabetic mother showing macrosomia and plethora. Born vaginally at 36 weeks' gestation, she weighed 5.5 kg and suffered a right-sided brachial plexus injury.

Neonatal problems include:

- *Hypoglycaemia*. Transient hypoglycaemia is common during the first day of life from fetal hyperinsulinism, but can often be prevented by early feeding. The infant's blood glucose should be closely monitored during the first 24 h and hypoglycaemia treated
- *Respiratory distress syndrome (RDS)*. More common as lung maturation is delayed
- *Hypertrophic cardiomyopathy*. Hypertrophy of the cardiac septum occurs in some infants. It regresses over several weeks but may cause heart failure from reduced left ventricular function
- *Polycythaemia* (venous haematocrit >0.65). Makes the infant look plethoric. Treatment with partial exchange transfusion to reduce the haematocrit and normalise viscosity may be required.

Gestational diabetes is when carbohydrate intolerance occurs only during pregnancy. Its definition and method of identification remain controversial. It is more common in women who are obese and in those of Afro-Caribbean and Asian ethnicity. The incidence of macrosomia and its complications is similar to that of the insulin-dependent diabetic mother, but the incidence of congenital malformations is not increased. However, there are an increasing number of mothers with type 2 non-insulin dependent diabetes, associated with the increase in obesity in the population. Their fetuses are also at increased risk of congenital malformations.

Summary

Maternal diabetes

- Meticulous control pre-conceptually and during pregnancy markedly reduces fetal and neonatal morbidity and mortality.
- The fetus may be macrosomic because of fetal hyperglycaemia resulting in hyperinsulinism, or growth-restricted secondary to maternal microvascular disease, and is at increased risk of congenital malformations.
- The macrosomic infant is at increased risk of asphyxia and birth trauma from obstructed labour or delivery.
- The newborn infant is prone to hypoglycaemia and polycythaemia.

Hyperthyroidism

If mothers have had Graves' disease, 1–2% of their newborn infants are hyperthyroid, due to circulating thyroid-stimulating antibody, which crosses the placenta and stimulates the fetal thyroid. Hyperthyroidism in the fetus is suggested by fetal tachycardia on the CTG trace, and fetal goitre may be evident on ultrasound; in the neonate it is suggested by irritability,

weight loss, tachycardia, heart failure, diarrhoea and exophthalmos. Treatment with anti-thyroid drugs may be necessary for several months until the condition resolves.

Systemic lupus erythematosus

Systemic lupus erythematosus (SLE) with antiphospholipid syndrome is associated with recurrent miscarriage, intrauterine growth restriction, pre-eclampsia, placental abruption and preterm delivery. Some of the infants born to mothers with antibodies to the Ro (SS-A) or La (SS-B) antigens develop neonatal lupus syndrome, in which there is a self-limiting rash and, rarely, heart block.

Autoimmune thrombocytopenic purpura

In maternal autoimmune thrombocytopenic purpura (AITP), the fetus may become thrombocytopenic because maternal IgG antibodies cross the placenta and damage fetal platelets. Severe fetal thrombocytopenia places the fetus at risk of intracranial haemorrhage following birth trauma. Infants with severe thrombocytopenia or petechiae at birth should be given intravenous immunoglobulin. Platelet transfusions may be required if there is acute bleeding.

Maternal drugs affecting the fetus

Relatively few drugs are known definitely to damage the fetus (Table 9.1), but it is clearly advisable for pregnant women to avoid taking medicines unless it is essential. While the teratogenicity of a drug may be recognised if it causes malformations which are severe and distinctive, as with limb shortening following thalidomide ingestion, milder and less distinctive abnormalities may go unrecognised.

The problem of establishing a link may be compounded by a delay of months or years before any problems present. An example of this is diethylstilboestrol (DES), given in the past for threatened miscarriage, and its subsequent association with vaginal adenosis and carcinoma of the vagina and cervix in female offspring during adolescence or early adult life.

Alcohol and smoking

Excessive alcohol ingestion during pregnancy is sometimes associated with the 'fetal alcohol syndrome'. Its clinical features are growth restriction, characteristic face (Fig. 9.5), developmental delay and cardiac defects (up to 70%). The effects of less severe ingestion and binge-drinking remain uncertain but may affect growth and development, and mothers are advised to avoid alcohol (Department of Health, London, UK). Maternal

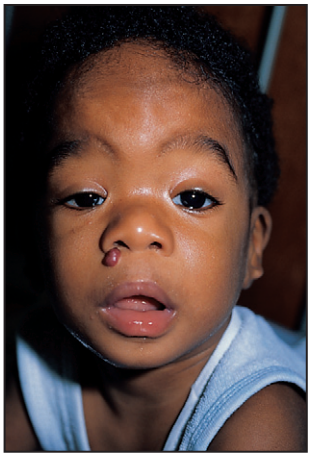


Figure 9.5 Characteristic facies of fetal alcohol syndrome with: a saddle-shaped nose; maxillary hypoplasia; absent philtrum between the nose and upper lip; and short, thin upper lip. This child also has a strawberry naevus below the right nostril.

Table 9.1 Maternal medication which may adversely affect the fetus

Medication	Adverse effect on fetus
Anticonvulsant therapy with carbamazepine, valproic acid (sodium valproate) or hydantoins (phenytoin)	Fetal carbamazepine/valproate/hydantoin syndrome – midfacial hypoplasia, CNS, limb and cardiac malformations, developmental delay
Cytotoxic agents	Congenital malformations
Diethylstilboestrol (DES)	Clear-cell adenocarcinoma of vagina and cervix
Iodides/propylthiouracil	Goitre, hypothyroidism
Lithium	Congenital heart disease
Tetracycline	Enamel hypoplasia of the teeth
Thalidomide	Limb shortening (phocomelia)
Vitamin A and retinoids	Increased spontaneous abortions, abnormal face
Warfarin	Interferes with cartilage formation (nasal hypoplasia and epiphyseal stippling); cerebral haemorrhages and microcephaly

cigarette smoking is associated in the fetus with an increased risk of miscarriage and stillbirth, a reduction in birthweight and IUGR (see pre-pregnancy care, earlier in this chapter).

Drug abuse

Maternal drug abuse with opiates is associated with an increased risk of prematurity and growth restriction. Many narcotic abusers take multiple drugs. Infants of mothers abusing heroin, methadone and other opiates during pregnancy often show evidence of drug withdrawal, with jitteriness, sneezing, yawning, poor feeding, vomiting, diarrhoea, weight loss and seizures during the first 2 weeks of life. Cocaine abuse is associated with placental abruption and preterm delivery, but rarely with withdrawal in the infant, although it may result in cerebral infarction. Amphetamine abuse is also associated with gastrointestinal and cerebral infarction. Mothers who abuse drugs, and their infants, are also at increased risk of hepatitis B and C and HIV infection. Hepatitis B vaccine is given to babies when indicated. Social and child protection aspects must be considered and Social Services involved.

Infants who develop significant features of drug withdrawal require admission to the Neonatal Unit and treatment. Oral morphine, methadone and diazepam are used at different centres. One of the major problems in managing these infants is that the parents' lifestyle and temperament are often not conducive to the needs of babies and young children. Close supervision or alternative caregivers are often required.

If there are unexplained clinical signs in an infant, consider drug withdrawal.

Drugs given during labour

Potential adverse effects to the fetus of drugs given during labour are:

- *Opioid analgesics/anaesthetic agents*. May suppress respiration at birth
- *Epidural anaesthesia*. May cause maternal pyrexia during labour. It is often difficult to differentiate this from fever caused by an infection. There is an increase in the rate of forceps deliveries
- *Sedatives, e.g. diazepam*. May cause sedation, hypothermia and hypotension in the newborn
- *Oxytocin and prostaglandin F2*. May cause hyperstimulation of the uterus leading to fetal hypoxia. It is also associated with a small increase in bilirubin levels in the neonate
- *Intravenous fluids*. May cause neonatal hyponatraemia unless they contain an adequate concentration of sodium.

Congenital infections

Intrauterine infection is usually from maternal primary infection during pregnancy. Those that can damage the fetus are:

- Rubella
- Cytomegalovirus (CMV)
- *Toxoplasma gondii*
- Parvovirus
- Varicella zoster
- Syphilis.

Congenital infections

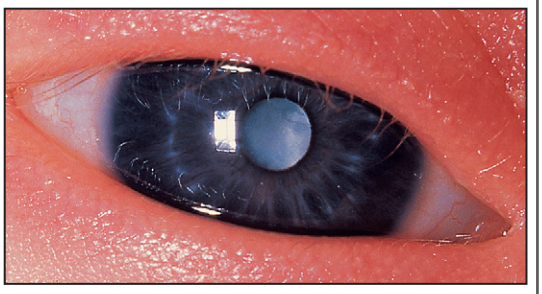


Figure 9.6a Cataract from congenital rubella. Congenital heart disease and deafness are the other common defects.

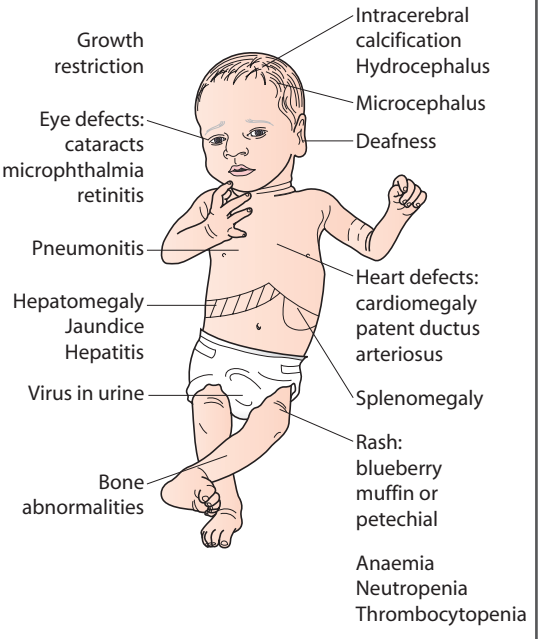


Figure 9.6b Clinical features of congenital rubella, cytomegalovirus (CMV), toxoplasmosis and syphilis.

Rubella

The diagnosis of maternal infection must be confirmed serologically as clinical diagnosis is unreliable. The risk and extent of fetal damage are mainly determined by the gestational age at the onset of maternal infection. Infection before 8 weeks' gestation causes deafness, congenital heart disease and cataracts in over 80% (Fig. 9.6a). About 30% of fetuses of mothers infected at 13–16 weeks' gestation have impaired hearing; beyond 18 weeks' gestation, the risk to the fetus is minimal.

Box 9.3 Diagnosis of congenital rubella, cytomegalovirus (CMV) and *Toxoplasma* infection

Mother	Seroconversion on screening serology
Fetus	Amniocentesis or chorionic villus sample, PCR
Placenta	Microscopy for syphilis, PCR
Urine from infant	Rubella, CMV – culture, PCR
Blood, CSF, other samples from infant	Culture, PCR
Blood serology	Rubella-specific IgM, CMV-specific IgM, *Toxoplasma*-specific IgM and persistently raised *Toxoplasma* IgG

Viraemia after birth continues to damage the infant. Tests used to confirm the diagnosis are shown in Box 9.3. The range of clinical features characteristic of congenital infections is shown in Figure 9.6b.

Congenital rubella is preventable. In the UK, it has become extremely rare since the measles/mumps/rubella (MMR) vaccine was introduced into the childhood immunisation programme, but this is dependent on the maintenance of a high vaccine uptake rate.

Cytomegalovirus

CMV is the most common congenital infection, affecting 3–4/1000 live births in the UK, with higher rates reported in parts of the USA. In Europe, 50% of pregnant women are susceptible to CMV. About 1% of susceptible women will have a primary infection during pregnancy, and in about 40% of them the infant becomes infected. The infant may also become infected following an episode of recurrent infection in the mother, but this is much less likely to damage the fetus. When an infant is infected:

- 90% are normal at birth and develop normally
- 5% have clinical features at birth, such as hepatosplenomegaly and petechiae (Fig. 9.6b), most of whom will have neurodevelopmental disabilities such as sensorineural hearing loss, cerebral palsy, epilepsy and cognitive impairment
- 5% develop problems later in life, mainly sensorineural hearing loss.

Infection in the pregnant woman is usually asymptomatic or causes a mild non-specific illness. There is no CMV vaccine and pregnant women are not screened for CMV. Antiviral therapy for infected infants with ganciclovir is under investigation in randomised controlled trials.

Toxoplasmosis

Acute infection with *Toxoplasma gondii*, a protozoan parasite, may result from the consumption of raw or undercooked meat and from contact with the faeces of recently infected cats. In the UK, fewer than 20% of pregnant women have had past infection, in contrast to 80% in France and Austria. Transplacental infection may occur during the parasitaemia of a primary infection, and about 40% of fetuses become infected. In the UK, the incidence of congenital infection is only about 0.1 per 1000 live births. Most infected infants are asymptomatic. About 10% have clinical manifestations (Fig. 9.6b), of which the most common are:

- Retinopathy, an acute fundal chorioretinitis which sometimes interferes with vision
- Cerebral calcification
- Hydrocephalus.

These infants usually have long-term neurological disabilities. Infected newborn infants are usually treated (pyrimethamine and sulfadiazine) for 1 year. Asymptomatic infants remain at risk of developing chorioretinitis into adulthood.

Varicella zoster

A total of 15% of pregnant women are susceptible to varicella (chickenpox). Usually, the fetus is unaffected but will be at risk if the mother develops chickenpox:

- in the first half of pregnancy (<20 weeks), when there is a <2% risk of the fetus developing severe scarring of the skin and possibly ocular and neurological damage and digital dysplasia
- within 5 days before or 2 days after delivery, when the fetus is unprotected by maternal antibodies and the viral dose is high. About 25% develop a vesicular rash. The illness has a mortality as high as 30%.

Exposed susceptible mothers can be protected with varicella zoster immune globulin (VZIG) and treated with aciclovir. Infants born in the high-risk period should also receive zoster immune globulin and are often also given aciclovir prophylactically.

If a mother develops chickenpox shortly before or after delivery, the infant needs protection from infection.

Syphilis

Congenital syphilis is rare in the UK. The clinical features are shown in Figure 9.6b. Those specific to congenital syphilis include a characteristic rash on the soles of the feet and hands and bone lesions. If mothers with syphilis identified on antenatal screening are fully treated 1 month or more before delivery, the infant does not require treatment and has an excellent prognosis. If there is any doubt about the adequacy of maternal treatment, the infant should be treated with penicillin.

Adaptation to extrauterine life

In the fetus, the lungs are filled with fluid, and oxygen is supplied by the placenta. The blood vessels that supply and drain the lungs are constricted (high

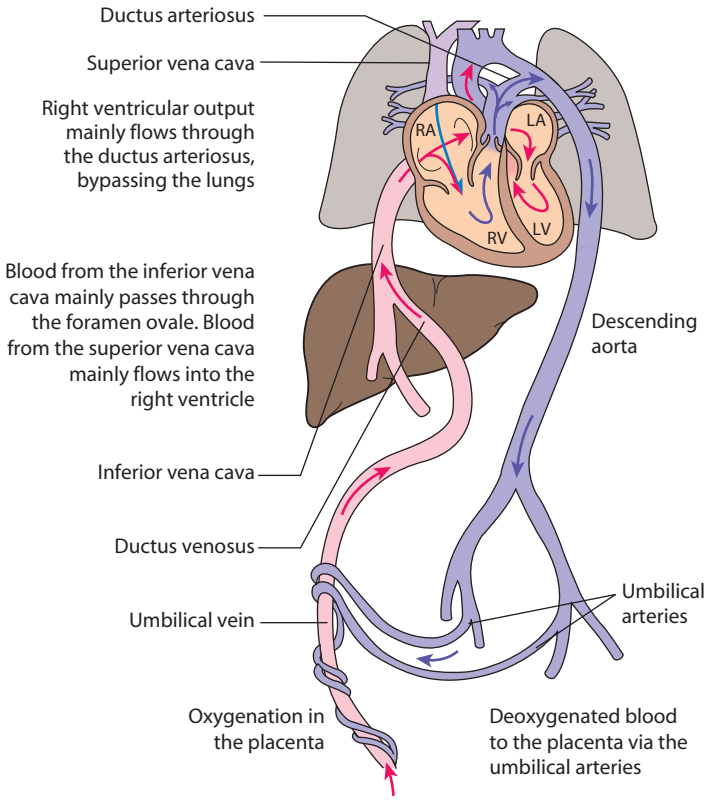


Figure 9.7 The fetal circulation.

pulmonary vascular resistance), so most blood from the right side of the heart bypasses the lungs and flows through the ductus arteriosus into the aorta, and some flows across the foramen ovale (Fig. 9.7). Shortly before and during labour, lung liquid production is reduced. During descent through the birth canal, the infant's chest is squeezed and some lung liquid drained. Multiple stimuli, including thermal, tactile and hormonal (with a particularly dramatic increase in catecholamine levels), initiate breathing. On average, the first breath occurs 6 s after delivery. Lung expansion is generated by intrathoracic negative pressure and a functional residual capacity is established. The mean time to establish regular breathing is 30 s. Once the infant gasps, the majority of the remaining lung fluid is absorbed into the lymphatic and pulmonary circulation.

Pulmonary expansion at birth is associated with a rise in oxygen tension, and with falling pulmonary vascular resistance the pulmonary blood flow increases. Increased left atrial filling results in a rise in the left atrial pressure with closure of the foramen ovale. The flow of oxygenated blood through the ductus arteriosus causes physiological, and eventual anatomical, ductal closure. After an elective caesarean section, when the mother has not been in labour and the infant's chest has not been squeezed through the birth canal, it may take several hours for the lung fluid to be completely absorbed, causing rapid, laboured breathing (transient tachypnoea of the newborn).

Some infants do not breathe at birth. This may be due to asphyxia, when the fetus experiences a lack of oxygen during labour and/or delivery. It does not necessarily mean that the brain has been injured but asphyxia can lead to brain injury or death. A fetus

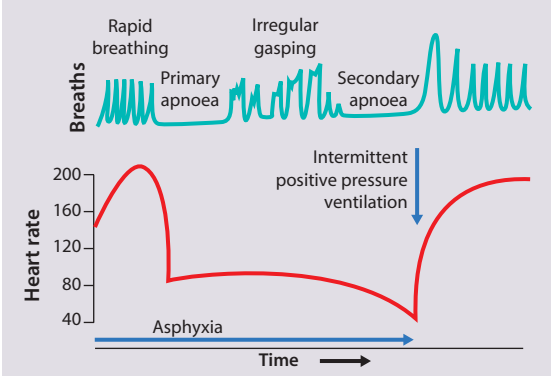


Figure 9.8 Changes in respiration and heart rate with continuous asphyxia. Once the infant has stopped gasping in secondary apnoea, resuscitation with lung expansion is required to establish regular respiration and restore the circulation.

deprived of oxygen in utero will attempt to breathe, but if this is unsuccessful (as it will be in utero), it will then become apnoeic (primary apnoea), during which time the heart rate is maintained. If oxygen deprivation continues, primary apnoea is followed by irregular gasping and then a second period of apnoea (secondary or terminal apnoea), when the heart rate and blood pressure fall. If delivered at this stage, the infant will only recover if help with lung expansion is provided, e.g. by positive pressure ventilation by mask or tracheal tube (Fig. 9.8).

The human fetus rarely experiences a continuous asphyxial insult, except after placental abruption or

Table 9.2 The Apgar score

	Score		
	0	**1**	**2**
Heart rate	Absent	<100 beats/min	≥100 beats/min
Respiratory effort	Absent	Gasping or irregular	Regular, strong cry
Muscle tone	Flaccid	Some flexion of limbs	Well flexed, active
Reflex irritability	None	Grimace	Cry, cough
Colour	Pale/blue	Body pink, extremities blue	Pink

complete occlusion of umbilical blood flow in a cord prolapse. More commonly, asphyxia, which occurs during labour and delivery is intermittent, e.g. from prolonged and frequent uterine contractions. Although birth asphyxia is an important cause of failure to establish breathing requiring resuscitation at birth, there are other causes, including birth trauma, maternal analgesic or anaesthetic agents, retained lung fluid, preterm infant or a congenital malformation which interferes with breathing.

The Apgar score is used to describe a baby's condition at 1 and 5 min after delivery (Table 9.2). It is also measured at 5-min intervals thereafter, if the infant's condition remains poor. The most important components are the heart rate and respiration.

Neonatal resuscitation

Most infants do not require any resuscitation. Shortly after birth, the baby will take a breath or cry, establish regular breathing and become pink. The baby can be handed directly to his or her mother, and covered with a warm towel to avoid becoming cold. However, a newborn infant who does not establish normal respiration directly will need to be transferred to a resuscitation table for further assessment (Fig. 9.9a). There should be an overhead radiant heater and the infant should be dried and partially covered and kept warm. Suction of the mouth and nose is normally unnecessary and vigorous suction of the back of the throat may provoke bradycardia from vagal stimulation. If the infant's breathing in the first minute of life is irregular or shallow, but the heart rate is satisfactory (>100 beats/min), breathing is encouraged with airway opening manoeuvres.

If the infant does not start to breathe, or if the heart rate drops below 100 beats/min, airway positioning and lung inflation by breathing by mask ventilation are started (Fig. 9.9b–f). If the baby's condition does not improve promptly, or if the infant is clearly in very poor condition at birth, additional assistance should be summoned immediately while continuing to maintain ventilation. If an attendant has the appropriate skills, tracheal intubation can be performed (Fig. 9.9g).

After tracheal intubation, if the heart rate does not increase and adequate chest movement is not achieved, consider 'DOPE':

- **D**isplaced tube: often in the oesophagus or right main bronchus
- **O**bstructed tube: especially meconium
- **P**atient:
  - tracheal obstruction
  - lung disorders: lung immaturity or respiratory distress syndrome, pneumothorax, diaphragmatic hernia, lung hypoplasia, pleural effusion
  - shock from blood loss
  - birth asphyxia or trauma
  - upper airways obstruction: choanal atresia.
- **E**quipment failure: gas supply exhausted or disconnected.

If there is any uncertainty about the adequacy of ventilation in an intubated baby, consider removing the tracheal tube, give mask ventilation and then re-intubate.

If at any time the heart rate drops below 60 beats/min, provided adequate breathing has been achieved, chest compressions should be given (Fig. 9.9h–j). If the response to ventilation and chest compression remains inadequate, drugs should be given (Fig. 9.9k). Evidence for their efficacy is very poor.

Providing lung inflation, evidenced by good chest wall movement, is the key to successful neonatal resuscitation.

Meconium aspiration

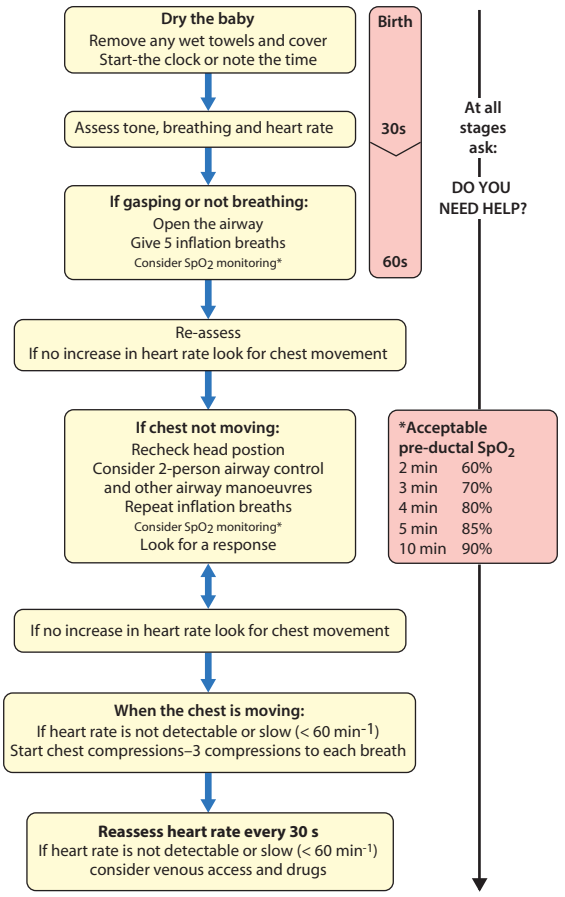
The passage of meconium becomes increasingly common the greater the infant's gestational age, particularly when post-term. Infants who also become acidotic may inhale thick meconium and develop meconium aspiration syndrome. Attempting to aspirate meconium from the nose and mouth while the infant's head is on the perineum is not recommended, as it is ineffective. If the infant cries at birth and establishes regular respiration, no resuscitation is required. If respiration is not established, the larynx should be inspected under direct vision and any thick meconium aspirated by suctioning with a large-bore suction catheter, but if the infant becomes bradycardic, positive pressure ventilation will be needed despite the presence of meconium.

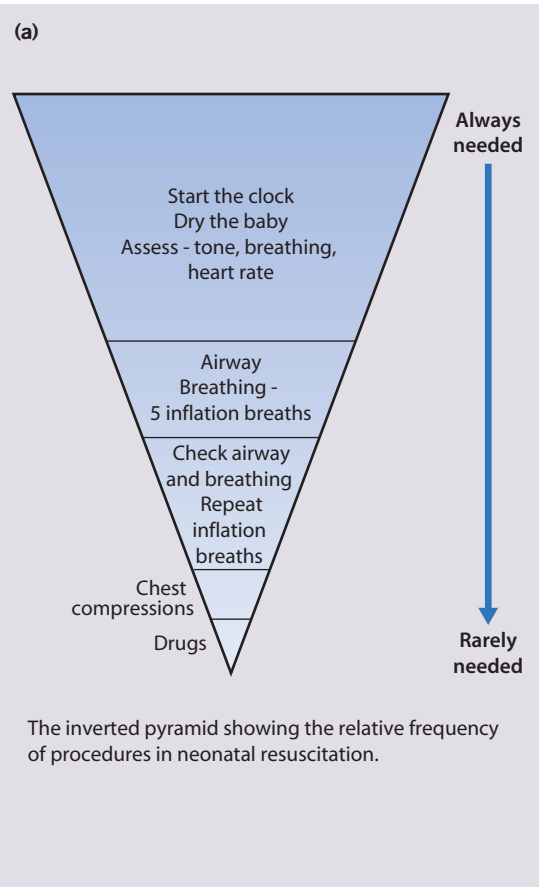
Neonatal resuscitation – Preparation and overview

Preparation

- All health professionals dealing with newborn infants should be proficient in basic resuscitation; i.e. **A**irway, **B**reathing with mask ventilation, **C**irculation with cardiac compressions
- Additional skilled assistance is needed if the baby does not respond rapidly and should be called without delay
- A person proficient in advanced resuscitation (**A**irway, **B**reathing via tracheal ventilation, **C**irculation, **D**rugs) should be on site and available at short notice in a maternity unit at all times
- The need for resuscitation can usually be anticipated and a person proficient in advanced resuscitation should be in attendance at all high-risk deliveries
- A clock should be started at birth for accurate timing of changes in the infant's condition
- Keep the infant warm. Dry, remove wet towel and replace with dry one. This will also provide stimulation. Can place directly on mother's chest and covered if crying, good tone and colour and desired by the mother
- Resuscitation should be performed under a radiant warmer
- If preterm and <30 weeks' gestation, to avoid heat loss, place the infant in a plastic bag without drying but under a radiant warmer and on a warming mattress. Leave the head exposed and cover with a woollen hat.
- Assess the infant's condition. Is the baby breathing or crying, good heart rate (120–160 beats/min, best assessed by listening with a stethoscope), good colour and muscle tone?
- If not, commence neonatal resuscitation

Overview





The inverted pyramid showing the relative frequency of procedures in neonatal resuscitation.

Figure 9.9 Neonatal resuscitation. (*Newborn Life Support,* 3rd ed. 2011, Resuscitation Council (UK). Reproduced with the kind permission of the Resuscitation Council (UK).)

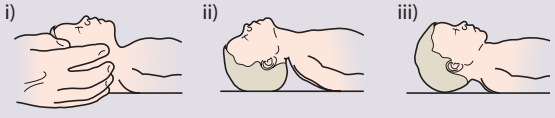
Airway and Breathing

Airway
- Opened by placing the infants's head in a neutral position **(b)**
- Provide chin lift or jaw thrust if necessary **(c)**
- Suction any blood or secretions
- Consider placing a Guedel airway

Breathing – mask ventilation
- If not breathing adequately, start mask ventilation
- Mask is placed over mouth and nose **(d)** and connected to flow-controlled pressure-limited circuit (e.g. mechanical ventilator or Neopuff) or self-inflating bag **(e)**
- Head in neutral position
- Give 5 inflation breaths, inflation time 2–3 seconds at inspiratory pressure of 30 cm H_2O in term infants to expand lungs
- If heart rate increases, but breathing does not start, continue with peak inspiratory pressure to achieve chest wall movement (15–25 cm H_2O, 0.5 second inflation time) and rate of 30–40 breaths/min
- Begin ventilatory resuscitation in air to avoid excessive tissue oxygenation. If giving additional oxygen, use air/oxygen blender to titrate oxygen concentration with oxygen saturation on pulse oximeter. (Acceptable pre-ductal saturations – 2 min 60%; 3 min 70%; 4 min 80%; 5 min 85%; 10 min 90%.)
- Reassess every 30 seconds. If heart rate not responding, check mask position, neck position, is jaw thrust needed, is circuit all right, ensure adequate chest movement. Consider using two-person airway control **(f)**. Call for help

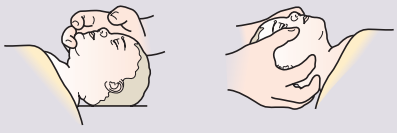
Intubation
- Intubation and mechanical ventilation **(g)** are indicated if: mask ventilation is ineffective, tracheal suction needed to clear an obstructed airway, congenital upper airway abnormality, extreme prematurity-for giving surfactant.
- Limit intubation attempts to 20–30 seconds.

(b) Head position, vital for airway management



i) Head in correct airway position ii) Head over-extended – incorrect
iii) Head flexed – incorrect

(c)



Chin support Jaw thrust

(d) Correct size and position of the face mask. It should cover the mouth, nose and chin

Correct
Covers mouth, nose and chin but not eyes



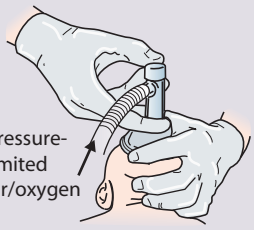
Incorrect
Too large – covers eyes and extends over chin



Incorrect
Too small – does not cover nose and mouth completely



(e) Mask ventilation



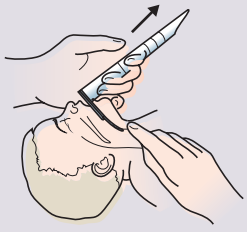
Mask ventilation delivered with pressure-limited circuit via T-piece (as shown), Neopuff or self-inflating bag.

(f) Two-person airway control



Consider if mask inflation ineffective. One person holds the head in the correct position, applies jaw thrust and holds the mask in place. The assistant operates the T-piece to provide lung inflation.

(g) Tracheal intubation



The laryngoscope blade is lifted upwards. Gentle pressure on the trachea helps bring the vocal cords into view

Figure 9.9, cont'd.

Neonatal resuscitation – Circulation and Drugs

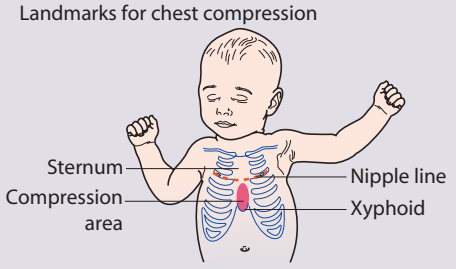
Circulation

Chest compression (h, i and j)

- Start if heart rate < 60 beats/min in spite of effective lung inflation
- Ratio of compression: lung inflation of 3:1, rate of 90 compressions: 30 breaths/min (120 events/min)
- Recheck heart rate every 30 seconds; stop when heart rate >60 beats/min

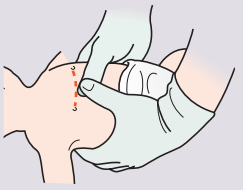
(h) Chest compression

Landmarks for chest compression



Apply pressure to lower third of sternum, just below imaginary line joining the nipples. Depress to reduce antero-posterior diameter by one-third (1–1.5 cm).

(i)



Thumb technique, with hands encircling the chest. In larger infants thumbs can be placed side by side.

(j)



Two finger technique – less effective but easier if alone.

Volume and drugs

Consider drugs **(k)** if heart rate <60 beats/min in spite of adequate ventilation and chest compression, though evidence for their efficacy is lacking
Rarely needed.
Drugs should be given via an umbilical venous catheter, or, if not possible, via an intra-osseous needle.
Drugs given via a peripheral vein are unlikely to reach the heart. Giving standard doses of epinephrine (adrenaline) down the endotracheal tube does not appear to be effective, so drug dosage is increased for this route.
A newborn baby who looks white and has poor skin and peripheral perfusion due to acidosis and peripheral vasoconstriction may have had acute blood loss. There may be a history of antepartum haemorrhage or acute twin-to-twin transfusion. Immediate blood transfusion with Group O rhesus negative blood is required.

(k) Drugs used in neonatal resuscitation

Drug	Concentration	Route/dosage	Indications
Epinephrine (adrenaline)	1:10 000	IV: 0.1 ml/kg (10 micrograms/kg), then 0.1– 0.3 ml/kg (10–30 micrograms/kg) ET: 1ml/kg (100 micrograms/kg) i.e. 10 times the IV dose, whilst IV access is obtained	Heart rate <60 beats/min in spite of adequate ventilation and external cardiac compression
Sodium bicarbonate	4.2%	2–4 ml/kg (1–2 mmol/kg)	Severe lactic acidosis
Dextrose	10%	2.5 ml/kg (250 mg/kg)	Hypoglycaemia
Volume expander	Normal saline Blood	10 ml/kg, repeat if necessary	Blood loss

Figure 9.9, cont'd.

Naloxone

Infants born to mothers who have received opiate analgesia within a few hours of delivery may occasionally develop respiratory depression, which can be reversed by naloxone. It is only given if respiration continues to be depressed following initial resuscitation. As the half-life of naloxone is shorter than that of the maternal opiate, the infant's breathing must be monitored for several hours, as further doses of naloxone may be required. With modern obstetric practice, naloxone is rarely needed. Naloxone should not be given to babies of opiate abusing mothers as acute withdrawal symptoms may be precipitated.

Resuscitation of the preterm infant

Preterm infants are particularly liable to hypothermia, and every effort must be made to keep them warm during resuscitation. Infants of <30 weeks' gestation

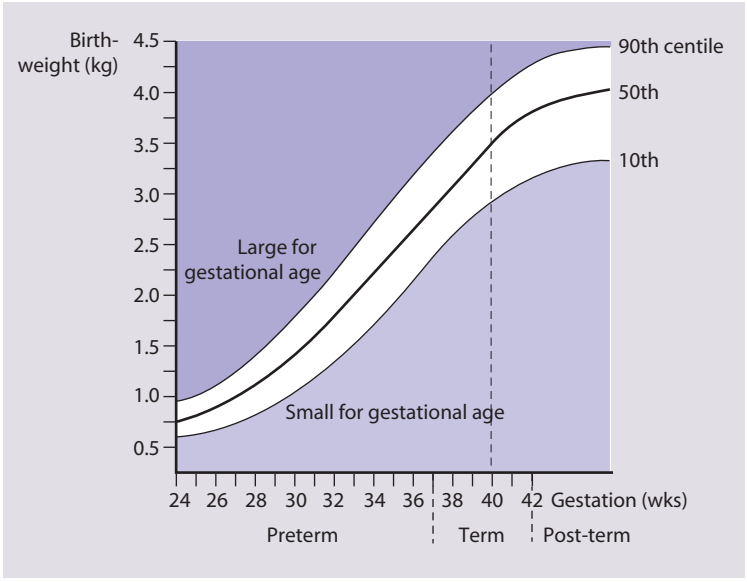


Figure 9.10 The birthweight of small-for-gestational-age infants is below the 10th centile for their gestation. Small-for-gestational-age infants may be preterm, term or post-term.

should, with the exception of the face, be placed into a plastic bag or wrapped in plastic sheeting. Excessive tissue oxygenation may cause tissue damage to the brain, lungs and eyes from oxygen free radicals. Ideally, an air/oxygen mixer should be used and any additional oxygen given titrated against oxygen saturation. Avoid exceeding a pre-ductal saturation of 95%. Very premature infants often develop respiratory distress syndrome, and early endotracheal administration of artificial surfactant may be indicated. Resuscitation of infants at the threshold of viability, at 22–24 weeks' gestation, raises particularly difficult ethical and management issues. An experienced paediatrician should be responsible for counselling the parents before delivery, if possible, and lead the management of the baby after birth.

Failure to respond to resuscitation

The decision to stop resuscitation is always difficult and should be made by a senior paediatrician. The longer it takes a baby to respond to resuscitation, the less likely is survival. If there is no breathing or cardiac output after 10 min of effective resuscitation, further efforts are likely to be fruitless and resuscitation should be stopped. If prolonged resuscitation has been required, the infant should be transferred to the neonatal unit for assessment and monitoring.

Size at birth

An infant's gestation and birthweight influence the nature of the medical problems likely to be encountered in the neonatal period. In the UK, 7% of babies are of low birthweight (<2.5 kg). However, they account for about 70% of neonatal deaths.

Definitions

Babies with a birthweight below the 10th centile for their gestational age are called small for gestational age or small-for-dates (Fig. 9.10). The majority of these infants are normal, but small. The incidence of congenital abnormalities and neonatal problems is higher in those whose birthweight falls below the second centile (approximately two standard deviations (SD) below the mean), and some authorities restrict the term to this group of babies. An infant's birthweight may also be low because of preterm birth, or because the infant is both preterm and small for gestational age.

Small-for-gestational-age infants may have grown normally but are small, or they may have experienced intrauterine growth restriction (IUGR), i.e. they have failed to reach their full genetically determined growth potential and appear thin and malnourished. Babies with a birthweight above the 10th centile may also be malnourished, e.g. a fetus growing along the 80th centile who develops growth failure and whose weight falls to the 20th centile.

Patterns of growth restriction

Growth restriction in both the fetus and infant has traditionally been classified as symmetrical or asymmetrical. In the more common asymmetrical growth restriction, the weight or abdominal circumference lies on a lower centile than that of the head. This occurs when the placenta fails to provide adequate nutrition late in pregnancy but brain growth is relatively spared at the expense of liver glycogen and skin fat (Fig. 9.11). This form of growth restriction is associated with uteroplacental dysfunction secondary to maternal pre-eclampsia, multiple pregnancy, maternal smoking or it may be idiopathic. These infants rapidly put on weight after birth.



Figure 9.11 Severe intrauterine growth restriction in a twin.

In symmetrical growth restriction, the head circumference is equally reduced. It suggests a prolonged period of poor intrauterine growth starting in early pregnancy (or that the gestational age is incorrect). It is usually due to a small but normal fetus, but may be due to a fetal chromosomal disorder or syndrome, a congenital infection, maternal drug and alcohol abuse or a chronic medical condition or malnutrition. These infants are more likely to remain small permanently.

In practice, distinction between asymmetrical and symmetrical growth restriction often cannot be made.

Monitoring the growth-restricted fetus

The fetus with IUGR is at risk from:

- intrauterine hypoxia and 'unexplained' intrauterine death
- asphyxia during labour and delivery.

The growth-restricted fetus will need to be monitored closely to determine the optimal time for delivery. Progressive uteroplacental failure results in:

- reduced growth in femur length and abdominal circumference
- abnormal umbilical artery Doppler waveforms – absent or reversed end diastolic flow velocity
- redistribution of blood flow in the fetus – increased to the brain, reduced to gastrointestinal tract, liver, skin and kidneys
- reduced amniotic fluid volume
- reduced fetal movements and abnormal CTG (cardiotocography).

The growth-restricted infant

After birth, these infants are liable to:

- hypothermia because of their relatively large surface area
- hypoglycaemia from poor fat and glycogen stores
- hypocalcaemia
- polycythaemia (venous haematocrit >0.65).

Large-for-gestational-age infants

Large-for-gestational-age infants are those above the 90th weight centile for their gestation. Macrosomia is a feature of infants of mothers with either permanent or gestational diabetes, or a baby with a congenital syndrome (e.g. Beckwith–Wiedemann syndrome). The problems associated with being large for gestational age are:

- Birth asphyxia from a difficult delivery
- Breathing difficulty from an enlarged tongue in Beckwith–Wiedemann syndrome
- Birth trauma, especially from shoulder dystocia at delivery (difficulty delivering the shoulders from impaction behind maternal symphysis pubis)
- Hypoglycaemia due to hyperinsulinism
- Polycythaemia.

Summary

Size at birth

- Small for gestational age – birthweight <10th centile
- Intrauterine growth restriction (IUGR) – fails to reach genetically determined growth potential
- Growth restriction – symmetrical or asymmetrical, but often mixed.

Routine examination of the newborn infant

Immediately after a baby is born, parents are naturally anxious to know if their baby is alright and appears normal. To answer this, the midwife (or the paediatrician or obstetrician, if present) will briefly but carefully check that the baby is pink, breathing normally and has no major abnormalities. If a significant problem is identified, an experienced paediatrician must explain the situation to the parents. If the baby is markedly preterm, small or ill, admission to a neonatal unit will be required. Should there be any uncertainty about the child's sex, it is important not to guess but to explain to the parents that further tests are necessary. Babies are given vitamin K at birth to prevent haemorrhagic disease of the newborn unless parents will not give consent.

Within 24 h of birth every baby should have a full and thorough examination, the 'routine examination of the newborn infant'. Its purpose is to:

- detect congenital abnormalities not already identified at birth, e.g. congenital heart disease, developmental dysplasia of the hip (DDH)
- check for potential problems arising from maternal disease or familial disorders

Routine examination of the newborn infant

Birthweight, gestational age and birthweight centile are noted.

General observation of the baby's appearance, posture and movements provides valuable information about many abnormalities. The baby must be fully undressed during the examination.

The head circumference is measured with a paper tape measure and its centile noted. This is a surrogate measure of brain size.

The fontanelle and sutures are palpated. The fontanelle size is very variable. The sagittal suture is often separated and the coronal sutures may be overriding. A tense fontanelle when the baby is not crying may be due to raised intracranial pressure and cranial ultrasound should be performed to check for hydrocephalus. A tense fontanelle is also a late sign of meningitis.

The face is observed. If abnormal, this may represent a syndrome, particularly if other anomalies are present. Down syndrome is the most common, but there are hundreds of syndromes. When the diagnosis is uncertain, a book or a computer database may be consulted and advice should be sought from a senior paediatrician or geneticist.



Figure 9.12 Term newborn. Median measurements:
- birthweight 3.5 kg;
- head circumference 35 cm;
- length 50 cm.

If plethoric or pale, the haematocrit should be checked to identify polycythaemia or anaemia. Central cyanosis, which always needs urgent assessment, is best seen on the tongue.

Jaundice within 24 h of birth requires further evaluation.

The eyes are checked for red reflex with an ophthalmoscope. If absent, may be from cataracts, retinoblastoma and corneal opacity. This reflex is not present in infants with pigmented skin, but the retinal vessels can be visualised.

The palate needs to be inspected, including posteriorly to exclude a posterior cleft palate, and palpated to detect an indentation of the posterior palate from a submucous cleft.

Breathing and chest wall movement are observed for signs of respiratory distress.

On auscultating the heart, the normal rate is 110–160 beats/min in term babies, but may drop to 85 beats/min during sleep.

On palpating the abdomen, the liver normally extends 1–2 cm below the costal margin, the spleen tip may be palpable, as may the kidney on the left side. Any intra-abdominal masses, which are usually renal in origin, need further investigation.

The femoral pulses are palpated. Their pulse pressure is:
– reduced in coarctation of the aorta. This can be confirmed by measuring the blood pressure in the arms and legs
– increased if there is a patent ductus arteriosus.

The genitalia and anus are inspected on removing the nappy. Patency of the anus is confirmed. In boys, the presence of testes in the scrotum is checked by palpation.

Muscle tone is assessed by observing limb movements. Most babies will support their head briefly when the trunk is held vertically.

The whole of the back and spine is observed, looking for any midline defects of the skin.

The hips are checked for developmental dysplasia of the hips (DDH). This is left until last as the procedure is uncomfortable.

Box 9.4 Routine examination of the newborn.

- provide an opportunity for the parents to discuss any questions about their baby.

Before approaching the mother and baby, the obstetric and neonatal notes must be checked to identify relevant information. The examination (Fig. 9.12a and b) should be performed with the mother or ideally both parents present. Many findings in the newborn resolve spontaneously (Box 9.5, Fig. 9.13). Common significant abnormalities detectable at birth are listed in Box 9.6. A serious congenital anomaly is present at birth in about 10–15/1000 live births (Table 9.3). In addition, many congenital anomalies, especially of the heart, present clinically at a later age.

Testing for developmental dysplasia of the hip (DDH)

To test for developmental dysplasia of the hip (DDH), also called congenital dislocation of the hip (CDH), the infant needs to be relaxed, as kicking or crying results

Lesions in newborn infants that resolve spontaneously

Box 9.5 Lesions in newborn infants that resolve spontaneously

Peripheral cyanosis of the hands and feet – common in the first day

Traumatic cyanosis from a cord round the baby's neck or from a face or brow presentation – causes blue discoloration of the skin, petechiae over the head and neck or affected part but not the tongue

Swollen eyelids and distortion of shape of the head from the delivery

Subconjunctival haemorrhages – occur during delivery

Small white pearls along the midline of the palate (Epstein pearls)

Cysts of the gums (epulis) or floor of the mouth (ranula)

Breast enlargement – may occur in newborn babies of either sex (Fig. 9.13a). A small amount of milk may be discharged

White vaginal discharge or small withdrawal bleed in girls. There may be a prolapse of a ring of vaginal mucosa

Capillary haemangioma or 'stork bites' – pink macules on the upper eyelids, mid-forehead and nape of the neck are common and arise from distension of the dermal capillaries. Those on the eyelids gradually fade over the first year; those on the neck become covered with hair

Neonatal urticaria (erythema toxicum) – a common rash appearing at 2–3 days of age, consisting of white pinpoint papules at the centre of an erythematous base (Fig. 9.13b). The fluid contains eosinophils. The lesions are concentrated on the trunk; they come and go at different sites

Milia – white pimples on the nose and cheeks, from retention of keratin and sebaceous material in the pilaceous follicles (Fig. 9.13c)

Mongolian blue spots – blue/black macular discoloration at the base of the spine and on the buttocks (Fig. 9.13d); occasionally occur on the legs and other parts of the body. Usually but not invariably in Afro-Caribbean or Asian infants. They fade slowly over the first few years. They are of no significance unless misdiagnosed as bruises

Umbilical hernia – common, particularly in Afro-Caribbean infants. No treatment is indicated as it usually resolves within the first 2–3 years

Positional talipes – the feet often remain in their in-utero position. Unlike true talipes equinovarus, the foot can be fully dorsiflexed to touch the front of the lower leg (Fig. 9.13e,f)

Caput succedaneum (see Fig. 10.6).



Figure 9.13a Breast enlargement in a newborn infant.



Figure 9.13b Erythema toxicum (neonatal urticaria), often has a raised pale centre. (Courtesy of Dr Nim Subhedar.)



Figure 9.13c Milia. (Courtesy of Dr Rodney Rivers.)



Figure 9.13d Mongolian blue spot.

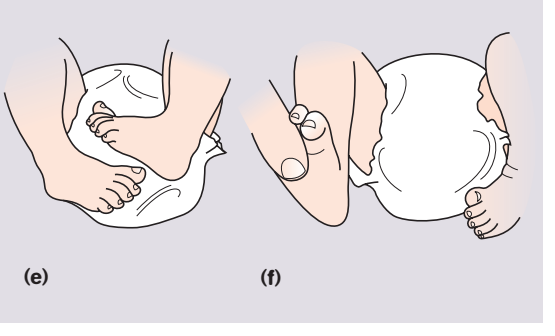


Figure 9.13e Positional talipes. Appearance at birth.
Figure 9.13f The foot can be fully dorsiflexed to touch the front of the lower leg. In true talipes equinovarus this is not possible.

Some significant abnormalities detected on routine examination

Box 9.6 Some significant abnormalities detected on routine examination

Port-wine stain (naevus flammeus). Present from birth and usually grows with the infant (Fig. 9.14a). It is due to a vascular malformation of the capillaries in the dermis. Rarely, if along the distribution of the trigeminal nerve, it may be associated with intracranial vascular anomalies (Sturge–Weber syndrome), or severe lesions on the limbs with bone hypertrophy (Klippel–Trenaunay syndrome). Disfiguring lesions can now be improved with laser therapy.

Strawberry naevus (cavernous haemangioma). Often not present at birth, but appear in the first month of life and may be multiple (Fig. 9.14b). It is more common in preterm infants. It increases in size until 3–15 months old, then gradually regresses. No treatment is indicated unless the lesion interferes with vision or the airway. Ulceration or haemorrhage may occur. Thrombocytopenia may occur with large lesions, when therapy with systemic steroids or interferon-α may be required.

Natal teeth consisting of the front lower incisors – may be present at birth. If loose, they should be removed to avoid the risk of aspiration.

Extra digits – are sometimes connected by a thin skin tag but may be completely attached containing bone (Fig. 9.14c) and should ideally be removed by a plastic surgeon or else tied off at its base. Skin tags anterior to the ear and accessory auricles should be removed by a plastic surgeon.

Heart murmur – poses a difficult problem, as most murmurs audible in the first few days of life resolve shortly afterwards. However, some are caused by congenital heart disease. If there are any features of a significant murmur (see Ch. 17), upper and lower limb blood pressures, and pre- and post-ductal pulse oximetry should be checked followed by an echocardiogram. Otherwise, a follow-up examination is arranged and the parents warned to seek medical assistance if their baby feeds poorly, develops laboured breathing or becomes cyanosed.

Midline abnormality over the spine or skull, such as a tuft of hair, swelling or naevus – requires further evaluation as it may indicate an underlying abnormality of the vertebrae, spinal cord or brain.

Palpable and large bladder – if there is urinary outflow obstruction, particularly in boys with posterior urethral valves. Requires prompt evaluation with ultrasound.

Talipes equinovarus – which cannot be corrected as in positional talipes.



Figure 9.14a Port-wine stain in an infant.



Figure 9.14b Strawberry naevus.



Figure 9.14c Extra digits.

Table 9.3 Prevalence of serious congenital anomalies per 1000 live births (England and Wales)

Anomaly	Prevalence
Congenital heart disease	6–8 (0.8 on the first day of life)
Developmental dysplasia of the hip	1.5 (but about 6/1000 have an abnormal initial clinical examination)
Talipes	1.0
Down syndrome	1.0
Cleft lip and palate	0.8
Urogenital (hypospadias, undescended testes)	1.2
Spina bifida/ anencephaly	0.1

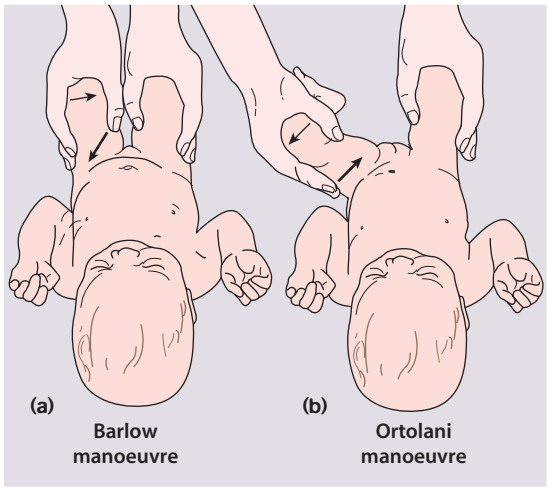


Figure 9.15 **(a)** The hip is dislocated posteriorly out of the acetabulum (Barlow manoeuvre). **(b)** The dislocated hip is relocated back into the acetabulum (Ortolani manoeuvre).

in tightening of the muscles around the hip and prevents satisfactory examination. The pelvis is stabilised with one hand. With the other hand, the examiner's middle finger is placed over the greater trochanter and the thumb around the distal medial femur. The hip is held flexed and adducted. The femoral head is gently pushed downwards. If the hip is dislocatable, the femoral head will be pushed posteriorly out of the acetabulum (Fig. 9.15a).

The next part of the examination is to see if the hip can be returned from its dislocated position back into the acetabulum. With the hip abducted, upward leverage is applied (Fig. 9.15b). A dislocated hip will return with a 'clunk' into the acetabulum. Ligamentous clicks without any movement of the head of femur are of no significance. It should also be possible to abduct the hips fully, but this may be restricted if the hip is dislocated. Clinical examination does not identify some infants who have hip dysplasia from lack of development of the acetabular shelf. DDH is more common in girls (six-fold increase), if there is a positive family history (20% of affected infants), if the birth is a breech presentation (30% of affected infants) or if the infant has a neuromuscular disorder.

Early recognition of DDH is important as early splinting in abduction reduces long-term morbidity. A specialist orthopaedic opinion should be sought in the management of this condition. Ultrasound examination of the hip joint is performed increasingly in many hospitals, either following an abnormal examination or to screen babies at increased risk (breech presentation or positive family history). Ultrasound examination can be performed to screen all babies, but is not currently recommended in the UK as it is expensive, requires considerable expertise and there are many false positives. It will, however, identify some babies missed on clinical examination.

Vitamin K therapy

Vitamin K deficiency may result in haemorrhagic disease of the newborn. This disorder can occur early, during the first week of life, or late, from 1 to 8 weeks of age. In most affected infants, the haemorrhage is mild, such as bruising, haematemesis and melaena, or prolonged bleeding of the umbilical stump or after a circumcision. However, some suffer from intracranial haemorrhage, half of whom are permanently disabled or die.

Breast milk is a poor source of vitamin K, whereas infant formula milk has a much higher vitamin K content. Haemorrhagic disease of the newborn may occur in infants who are wholly breast-fed but not if fed with an infant formula. Infants of mothers taking anticonvulsants, which impair the synthesis of vitamin K-dependent clotting factors, are at increased risk of haemorrhagic disease, both during delivery and soon after birth. Infants with liver disease are also at increased risk.

The disease can be prevented if vitamin K is given by intramuscular injection, and in the UK, it was widely given to all newborn infants immediately after birth. In the early 1990s, one study suggested a possible association between vitamin K given intramuscularly and the development of cancer in childhood, but this has not been found in other, much larger studies. It is still recommended that all newborn infants are given intramuscular vitamin K. However, parents may request oral vitamin K as an alternative. As absorption via the oral route is variable, three doses are needed over the first 4 weeks of life to achieve adequate liver storage. Mothers on anticonvulsant therapy should receive oral prophylaxis from 36 weeks' gestation and the baby should be given intramuscular vitamin K.

Vitamin K should be given to all newborn infants to prevent haemorrhagic disease of the newborn.

Biochemical screening (Guthrie test)

Biochemical screening is performed on every baby. A blood sample, usually a heel prick, is taken when feeding has been established on day 5–9 of life. In the UK, all infants are screened for:

- phenylketonuria
- hypothyroidism
- haemoglobinopathies (sickle cell and thalassaemia)
- cystic fibrosis
- MCAD (medium-chain acyl-CoA dehydrogenase) deficiency – a rare inborn error of mitochondrial fatty acid metabolism causing acute illness and hypoglycaemia following fasting, which may also present as an ALTE (acute life-threatening episode).

Screening for cystic fibrosis is performed by measuring the serum immunoreactive trypsin, which is raised if there is pancreatic duct obstruction. If raised, DNA analysis is also performed to reduce the false-positive rate (see Ch. 16).

In the UK, biochemical screening is performed on all babies to identify phenylketonuria, congenital hypothyroidism, haemoglobinopathies, cystic fibrosis and MCAD deficiency.

Newborn hearing screening

Universal screening has been introduced in the UK to detect severe hearing impairment in newborn infants. Early detection and intervention improves speech and language. Evoked otoacoustic emission (EOAE) testing, when an earphone is placed over the ear and a sound is emitted which evokes an echo or emission from the ear if cochlear function is normal, is used as the initial screening test. If a normal test is not achieved, testing with automated auditory brainstem response (AABR) audiometry, using computer analysis of EEG waveforms evoked in response to a series of clicks, is performed, with referral to a paediatric audiologist if abnormal (see Ch. 3 for further details).

Newborn hearing screening is performed on all infants to detect severe hearing impairment.

Further reading

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10

Neonatal medicine

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The dramatic reduction in neonatal mortality throughout the developed world has resulted from advances in the management of newborn infants together with improvements in maternal health and obstetric care. Neonatal intensive care became increasingly available in the UK from 1975, and it is since that time that the mortality of very low birthweight (VLBW) infants has fallen (Fig. 10.1).

About 8–10% of babies born in the UK are admitted to a neonatal unit for special medical and nursing care, although whenever possible babies are cared for on postnatal wards to avoid separating mothers from their babies. About 1–3% of babies require intensive care.

In the UK, neonatal units are organised as networks, with units providing either:

- special care (level 1)
- short-term intensive care (level 2)
- long-term intensive care (level 3), usually linked to specialist fetal and obstetric care to form a specialist tertiary perinatal centre.

Modern technology allows even tiny preterm infants to benefit from the full range of intensive care, anaesthesia and surgery. If it is anticipated during pregnancy that the infant is likely to require long-term intensive care or surgery, it is preferable for the transfer to the

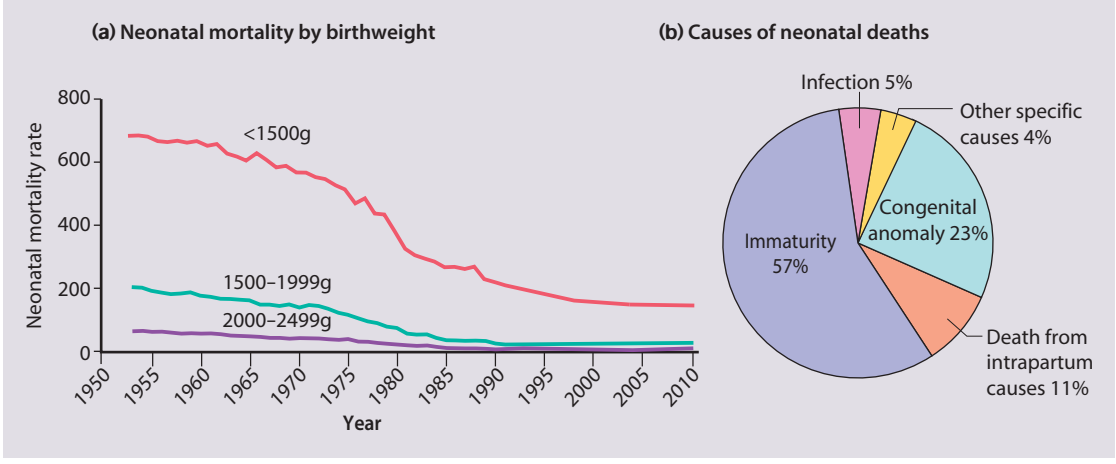


Figure 10.1 **(a)** The dramatic fall in neonatal mortality rate in England and Wales in the second half of the twentieth century, according to birthweight. In very low birthweight infants (<1500 g), the marked fall in the mortality occurred after the introduction of intensive care. **(b)** Causes of neonatal deaths in England and Wales, 2008. (Adapted from the Audit Commission and ONS 2010).

tertiary centre to be made in utero. When a baby requires transfer postnatally, transport should be by an experienced team of doctors and nurses. Arrangements should also be made for parents to be close to their infant during this stressful time.

Hypoxic-ischaemic encephalopathy

In perinatal asphyxia, gas exchange, either placental or pulmonary, is compromised or ceases altogether, resulting in cardiorespiratory depression. Hypoxia, hypercarbia and metabolic acidosis follow. Compromised cardiac output diminishes tissue perfusion, causing hypoxic-ischaemic injury to the brain and other organs. The neonatal condition is called hypoxic-ischaemic encephalopathy (HIE). It remains an important cause of brain damage, resulting in disability (Fig. 10.2) or death, and its prevention is one of the key aims of modern obstetric care. In developed countries, approximately 0.5–1/1000 liveborn term infants develop HIE and 0.3/1000 have significant neurologic disability. The incidence is higher in developing countries.

Most cases of hypoxic-ischaemic encephalopathy (HIE) follow a significant hypoxic event immediately before or during labour or delivery. These include:

- Failure of gas exchange across the placenta – excessive or prolonged uterine contractions, placental abruption, ruptured uterus
- Interruption of umbilical blood flow – cord compression including shoulder dystocia, cord prolapse
- Inadequate maternal placental perfusion, maternal hypotension or hypertension – often with intrauterine growth restriction
- Compromised fetus – anemia, intrauterine growth restriction
- Failure of cardiorespiratory adaptation at birth – failure to breathe.



Figure 10.2 Brain damage from severe birth asphyxia.

The clinical manifestations start immediately or up to 48 h after asphyxia, and can be graded:

- Mild – the infant is irritable, responds excessively to stimulation, may have staring of the eyes and hyperventilation and has impaired feeding
- Moderate – the infant shows marked abnormalities of tone and movement, cannot feed and may have seizures
- Severe – there are no normal spontaneous movements or response to pain; tone in the limbs may fluctuate between hypotonia and hypertonia; seizures are prolonged and often refractory to treatment; multi-organ failure is present.

The neuronal damage may be immediate from primary neuronal death or may be delayed from reperfusion injury causing secondary neuronal death. This delay offers the opportunity for neuroprotection with mild hypothermia.

Management

Skilled resuscitation and stabilisation of sick infants will minimise neuronal damage. Infants with HIE may need:

- respiratory support
- recording of amplitude-integrated electroencephalogram (aEEG, cerebral function monitor) to detect abnormal background activity to confirm early encephalopathy or identify seizures
- treatment of clinical seizures with anticonvulsants
- fluid restriction because of transient renal impairment
- treatment of hypotension by volume and inotrope support
- monitoring and treatment of hypoglycaemia and electrolyte imbalance, especially hypocalcaemia.

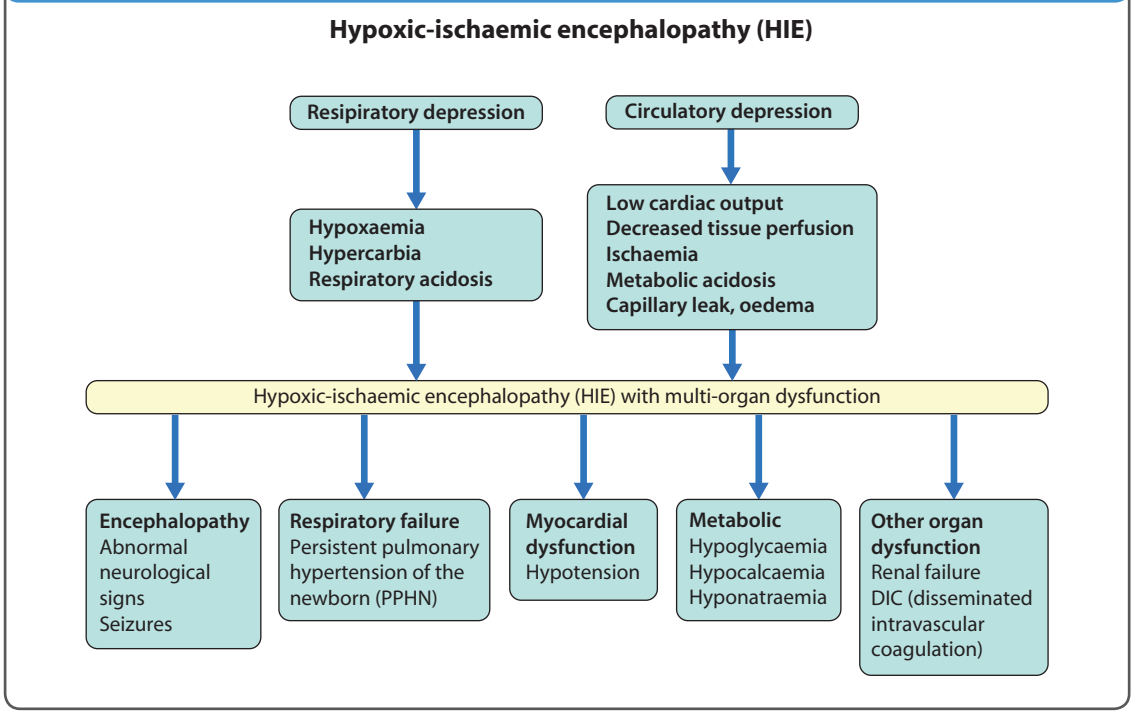
Recent randomised clinical trials have shown that mild hypothermia (cooling to a rectal temperature of 33–34°C for 72 h by wrapping the infant in a cooling blanket) reduces brain damage if started within 6 h of birth.

Prognosis

When HIE is mild, complete recovery can be expected. Infants with moderate HIE who have recovered fully on clinical neurological examination and are feeding normally by 2 weeks of age have an excellent long-term prognosis, but if clinical abnormalities persist beyond that time, full recovery is unlikely. Severe HIE has a mortality of 30–40%, and, of the survivors, over 80% have neurodevelopmental disabilities, particularly cerebral palsy. If magnetic resonance imaging (MRI) at 4–14 days in a term infant shows significant abnormalities (bilateral abnormalities in the basal ganglia and thalamus and lack of myelin in the posterior limb of the internal capsule), there is a very high risk of later cerebral palsy (Fig. 10.3).

Although hypoxic-ischaemic injury usually occurs antenatally or during labour or delivery, it may occur postnatally or be caused by a neonatal condition (e.g. inborn error of metabolism or kernicterus). The diagnosis 'birth asphyxia' has potentially serious medicolegal implications; it has been recommended that infants

Summary



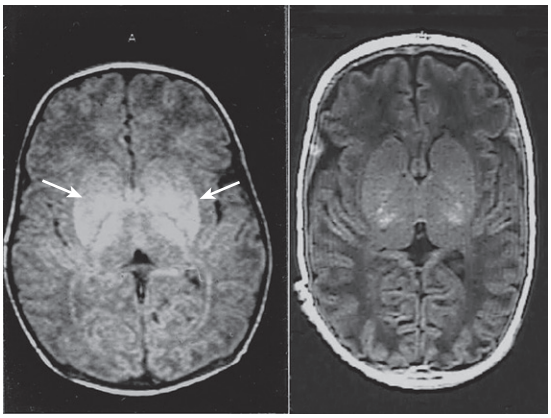


Figure 10.3 Magnetic resonance image of the brain at term. Left: hypoxic-ischaemic encephalopathy (HIE) showing abnormal (white) signal in the basal ganglia and thalami (arrows) and absence of signal in the internal capsule bilaterally. Right: normal scan showing grey basal ganglia and a white signal from myelin in the posterior limb of the internal capsule.

who have the clinical features of HIE should only be considered to have birth asphyxia if there is:

- evidence of severe hypoxia antenatally or during labour or at delivery
- resuscitation needed at birth
- features of encephalopathy
- evidence of hypoxic damage to other organs such as liver, kidney, or heart
- no other prenatal or postnatal cause identified
- characteristic findings on MRI neuroimaging.

Mild hypothermia for moderate and severe HIE reduces death and severe disability and increases the likelihood of survival with normal neurological function.

Birth injuries

Infants may be injured at birth, particularly if they are malpositioned or too large for the pelvic outlet. Injuries may also occur during manual manoeuvres, from forceps blades or at Ventouse deliveries. Fortunately, now that Caesarean section is available in every maternity unit, heroic attempts to achieve a vaginal delivery with resultant severe injuries to the infant have become extremely rare.

Soft tissue injuries

These include:

- Caput succedaneum (Fig. 10.4) – bruising and oedema of the presenting part extending beyond the margins of the skull bones; resolves in a few days
- Cephalhaematoma (Figs 10.4, 10.5) – haematoma from bleeding below the periosteum, confined within the margins of the skull sutures. It usually involves the parietal bone. The centre of the haematoma feels soft. It resolves over several weeks
- Chignon (Fig. 10.6) – oedema and bruising from Ventouse delivery
- Bruising to the face after a face presentation and to the genitalia and buttocks after breech delivery.

Birth injuries

Soft tissue injuries

- Caput succedaneum, cephalhaematoma, chignon, bruises and abrasions
- Subaponeurotic haemorrhage

Nerve palsies

- Brachial plexus – Erb palsy
- Facial nerve palsy

Fractures

- Clavicle, humerus, femur

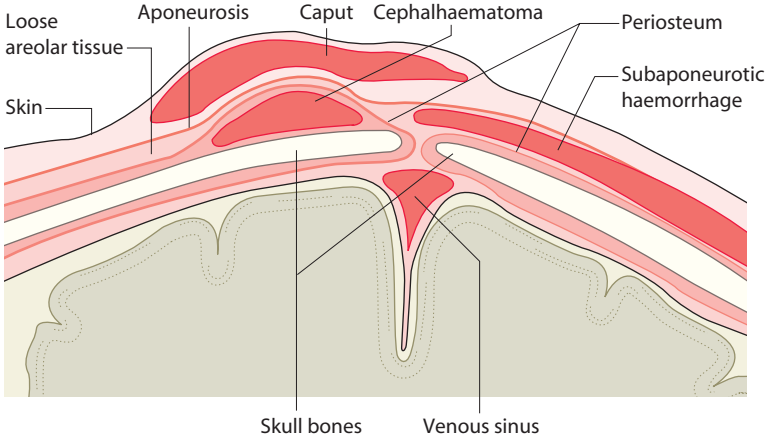


Figure 10.4 Location of extracranial haemorrhages.



Figure 10.5 A large cephalhaematoma.



Figure 10.6 Chignon.



Figure 10.7 Erb palsy. The affected arm lies straight, limp and with the hand pronated and the fingers flexed (waiter's tip position).

Preterm infants bruise readily from even mild trauma

- Abrasions to the skin from scalp electrodes applied during labour or from accidental scalpel incision at Caesarean section
- Forceps marks to face from pressure of blades – transient
- Subaponeurotic haemorrhage (Fig. 10.4) (very uncommon) – diffuse, boggy swelling of scalp on examination, blood loss may be severe and lead to hypovolaemic shock and coagulopathy.

Nerve palsies

Brachial nerve palsy results from traction to the brachial plexus nerve roots. They may occur at breech deliveries or with shoulder dystocia. Upper nerve root (C5 and C6) injury results in an Erb palsy (Fig. 10.7). It

may be accompanied by phrenic nerve palsy causing an elevated diaphragm. Most palsies resolve completely, but should be referred to an orthopaedic or plastic surgeon if not resolved by 2–3 months. Most recover by 2 years. A facial nerve palsy may result from compression of the facial nerve against the mother's ischial spine. It is unilateral, and there is facial weakness on crying but the eye remains open. It is usually transient, but methylcellulose drops may be needed for the eye. Rarely, nerve palsies may be from damage to the cervical spine, when there is lack of movement below the level of the lesion.

Fractures

Clavicle

Usually from shoulder dystocia. A snap may be heard at delivery or the infant may have reduced arm movement on the affected side, or a lump from callus formation may be noticed over the clavicle at several days of age. The prognosis is excellent and no specific treatment is required.

Humerus/femur

Usually mid-shaft, occurring at breech deliveries, or fracture of the humerus at shoulder dystocia. There is deformity, reduced movement of the limb and pain on movement. They heal rapidly with immobilisation.

Stabilising the preterm or sick infant

Preterm infants of <34 weeks' gestation and newborn infants who become seriously ill require their condition to be stabilised and monitored (Fig. 10.8). Many of them will need respiratory and circulatory support.

The preterm infant

The appearance, the likely clinical course, chances of survival and long-term prognosis depend on the gestational age at birth. The appearance and maturational changes of very preterm infants are described in Table 10.1 and the importance of parental involvement shown in Figures 10.10a and b. The external appearance and neurological findings can be scored to provide an estimate of an infant's gestational age (see Appendix).

The rate and severity of problems associated with prematurity decline markedly with increasing gestation. Infants born at 23–26 weeks' gestation encounter many problems (Box 10.1), require many weeks of intensive and special care in hospital and have a high overall mortality. With modern intensive care, the prognosis is excellent after 32 weeks' gestational age. The severity of an infant's respiratory disease and of any episodes of infection largely determine the neonatal course and outcome.

Box 10.1 Medical problems of preterm infants

- Need for resuscitation at birth
- Respiratory
  - Respiratory distress syndrome (RDS)
  - Pneumothorax
  - Apnoea and bradycardia
- Hypotension
- Patent ductus arteriosus
- Temperature control
- Metabolic
  - Hypoglycaemia
  - Hypocalcaemia
  - Electrolyte imbalance
  - Osteopenia of prematurity
- Nutrition
- Infection
- Jaundice
- Intraventricular haemorrhage/periventricular leukomalacia
- Necrotising enterocolitis
- Retinopathy of prematurity
- Anaemia of prematurity
- Iatrogenic
- Bronchopulmonary dysplasia (chronic lung disease)
- Inguinal hernias.

Respiratory distress syndrome

In respiratory distress syndrome (RDS), (also called hyaline membrane disease), there is a deficiency of surfactant, which lowers surface tension. Surfactant is a mixture of phospholipids and proteins excreted by the type II pneumocytes of the alveolar epithelium. Surfactant deficiency leads to widespread alveolar collapse and inadequate gas exchange. The more preterm the infant, the higher the incidence of RDS; it is common in infants born before 28 weeks' gestation and tends to be more severe in boys than girls. Surfactant deficiency is rare at term but may occur in infants of diabetic mothers. The term hyaline membrane disease derives from a proteinaceous exudate seen in the airways on histology. Glucocorticoids, given antenatally to the mother, stimulate fetal surfactant production and are given if preterm delivery is anticipated (see Ch. 9.)

The development of surfactant therapy has been a major advance. The preparations are natural, derived from extracts of calf or pig lung. They are instilled directly into the lung via a tracheal tube. Multinational placebo-controlled trials show that surfactant treatment reduces mortality from RDS by about 40%, without increasing the morbidity rate (Fig. 10.11).

At delivery or within 4 h of birth, babies with RDS develop clinical signs of:

- tachypnoea >60 breaths/min
- laboured breathing with chest wall recession (particularly sternal and subcostal indrawing) and nasal flaring

Stabilising preterm or sick infants

Airway, breathing

- Respiratory distress: tachypnoea, laboured breathing with chest wall recession, nasal flaring, expiratory grunting, cyanosis
- Apnoea

Management, as required:

- Clear the airway
- Oxygen
- High-flow humidified oxygen therapy
- CPAP (continuous positive airway pressure)
- Mechanical ventilation

Monitoring

- Oxygen saturation (maintain at 88–95% if preterm)
- Heart rate
- Respiratory rate
- Temperature
- Blood pressure
- Blood glucose
- Blood gases
- Weight

Temperature control

- Place in plastic bag at birth to keep warm if extremely preterm
- Perform stabilisation under a radiant warmer or in a humidified incubator to avoid hypothermia.

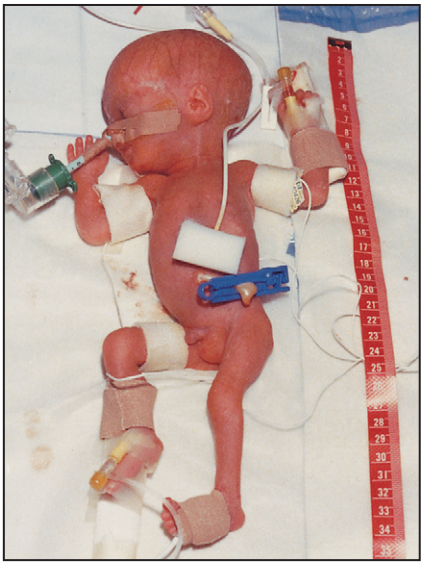


Figure 10.8 Stabilising preterm or sick infants is important to prevent complications. This preterm infant has leads on his limbs for monitoring heart rate and respiratory rate, temperature and oxygen saturation. There are arterial and intravenous cannulae and a nasotracheal tube for artificial ventilation.

Venous and arterial lines

Peripheral intravenous line

Required for intravenous fluids, antibiotics and other drugs.

Umbilical venous catheter

May be used for intravenous access at resuscitation, in extremely preterm infants for the first few days or to administer high osmolality fluids (e.g. high-concentration dextrose) or medications needing central delivery (e.g. inotropes).

Arterial line

- Inserted if frequent blood gas analysis, blood tests and continuous blood pressure monitoring are required. Usually umbilical artery catheter (UAC), sometimes peripheral cannula if for short period or no umbilical artery catheter possible.
- The arterial oxygen tension is maintained at 8–12 kPa (60–90 mmHg) and the CO_2 tension at 4.5–6.5 kPa (35–50 mmHg). Continuous non-invasive transcutaneous arterial blood gas monitoring may also be used.

Central venous line for parenteral nutrition, if indicated

Inserted peripherally when infant is stable.

Chest X-ray with or without abdominal X-ray

Assists in the diagnosis of respiratory disorders and to confirm the position of the tracheal tube and central lines.

Investigations

- Haemoglobin, neutrophil count, platelet count
- Blood urea, creatinine, electrolytes and lactate
- Culture – blood ± CSF ± urine
- Blood glucose
- CRP/acute phase reactant
- Coagulation screen if indicated

Antibiotics

Broad-spectrum antibiotics are given.

Minimal handling

All procedures, especially painful ones, adversely affect oxygenation and the circulation. Handling the infant is kept to a minimum and done as gently, rapidly and efficiently as possible. Analgesia should be provided to prevent pain as necessary.

Parents

Although medical and nursing staff are usually fully occupied stabilising the baby, time must be found for parents and immediate relatives to allow them to see and touch their baby and to be kept fully informed.

Figure 10.9
(a) Preterm infant.
(b) Term infant.



(a)



(b)

Table 10.1 The preterm infant compared with the term infant

Gestation	23–27 weeks	Term (37–42 weeks)
Birthweight (50th centile)	At 24 weeks – male 700 g, female 620 g	At 40 weeks – male 3.55 kg, female 3.4 kg
Skin	Very thin (Fig. 10.9a)	Thick skin (Fig. 10.9b)
	Dark red colour all over body	Pale pink colour
Ears	Pinna soft, no recoil	Pinna firm, cartilage to edge, immediate recoil
Breast tissue	No breast tissue palpable	One or both nodules >1 cm
Genitalia	Male – scrotum smooth, no testes in scrotum	Male – scrotum has rugae, testes in scrotum
	Female – prominent clitoris, labia majora widely separated, labia minora protruding	Female – labia minora and clitoris covered
Breathing	Needs respiratory support. Apnoea common	Rarely needs respiratory support. Apnoea rare
Sucking and swallowing	No coordinated sucking	Coordinated (from 34–35 weeks)
Feeding	Usually needs TPN (total parenteral nutrition), then tube feeding	Cries when hungry. Feeds on demand
Cry	Faint	Loud
Vision, interaction	Eyelids may be fused. Infrequent eye movements. Not available for interaction	Makes eye contact, alert wakefulness
Hearing	Startles to loud noise	Responds to sound
Posture	Limbs extended, jerky movements	Flexed posture, smooth movements



(a)



(b)

Figure 10.10a Parental involvement in neonatal care. Skin-to-skin contact between infant and parent (Kangaroo care) promotes bonding.
Figure 10.10b Parental involvement in neonatal care. Mother giving her baby expressed breast milk (in syringe) via nasogastric tube, allowing close eye and skin contact between mother and baby.

Respiratory distress syndrome

- Common in very preterm infants
- Caused by surfactant deficiency
- Antenatal corticosteroids and surfactant therapy markedly reduce morbidity and mortality.

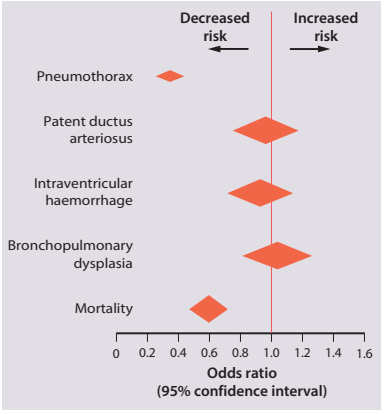


Figure 10.11 Meta-analysis of treatment of preterm infants with natural surfactant, showing a dramatic reduction in pneumothoraces and mortality.

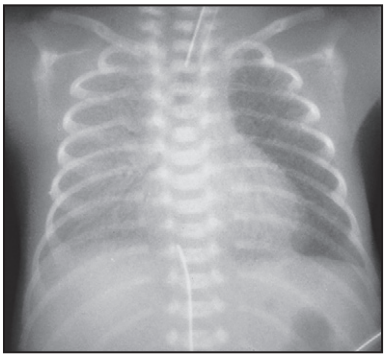


Figure 10.12 Chest X-ray in respiratory distress syndrome showing a diffuse granular or 'ground glass' appearance of the lungs and an air bronchogram, where the larger airways are outlined. The heart border becomes indistinct or obscured completely with severe disease. A tracheal tube and an umbilical artery catheter are present.

- expiratory grunting in order to try to create positive airway pressure during expiration and maintain functional residual capacity
- cyanosis if severe.

The characteristic chest X-ray appearance is shown in Figure 10.12. Treatment with raised ambient oxygen is required, which may need to be supplemented with continuous positive airway pressure (delivered via nasal cannulae) or artificial ventilation via a tracheal tube. The ventilatory requirements need to be adjusted according to the infant's oxygenation (which is measured continuously), chest wall movements and blood gas analyses. Mechanical ventilation (with intermittent positive pressure ventilation or high-frequency oscillation) may be required. High-flow humidified oxygen therapy, via nasal cannulae, may be used to wean babies from added oxygen therapy.

Surfactant therapy reduces morbidity and mortality of preterm infants with respiratory distress syndrome.

Pneumothorax

In respiratory distress syndrome, air from the overdistended alveoli may track into the interstitium, resulting in pulmonary interstitial emphysema (PIE). In up to 10% of infants ventilated for RDS, air leaks into the pleural cavity and causes a pneumothorax (Fig. 10.13). When this occurs, the infant's oxygen requirement usually increases, and the breath sounds and chest movement on the affected side are reduced, although this can be difficult to detect clinically. A pneumothorax may be demonstrated by transillumination with a bright fibreoptic light source applied to the chest wall. A tension pneumothorax is treated by inserting a chest drain. In order to try and prevent pneumothoraces, infants are ventilated with the lowest pressures that provide adequate chest movement and satisfactory blood gases.

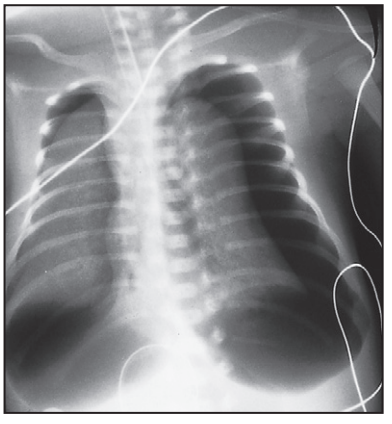


Figure 10.13 Chest X-ray showing bilateral pneumothoraces in a preterm infant with respiratory distress syndrome.

Apnoea and bradycardia and desaturation

Episodes of apnoea and bradycardia and desaturation are common in very low birthweight infants until they reach about 32 weeks' gestational age. Bradycardia

may occur either when an infant stops breathing for over 20–30 s or when breathing continues but against a closed glottis. An underlying cause (hypoxia, infection, anaemia, electrolyte disturbance, hypoglycaemia, seizures, heart failure or aspiration due to gastro-oesophageal reflux) needs to be excluded, but in many instances, the cause is immaturity of central respiratory control. Breathing will usually start again after gentle physical stimulation. Treatment with the respiratory stimulant caffeine often helps. Continuous positive airways pressure (CPAP) may be necessary if apnoeic episodes are frequent.

Temperature control

Hypothermia causes increased energy consumption and may result in hypoxia and hypoglycaemia, failure to gain weight and increased mortality. Preterm infants are particularly vulnerable to hypothermia, as:

- they have a large surface area relative to their mass, so there is greater heat loss (related to surface area) than heat generation (related to mass)
- their skin is thin and heat permeable, so transepidermal water loss is important in the first week of life
- they have little subcutaneous fat for insulation
- they are often nursed naked and cannot conserve heat by curling up or generate heat by shivering.

There is a neutral temperature range in which an infant's energy consumption is at a minimum level. In the very immature baby, this neutral temperature is highest during the first few days of life and subsequently declines. The temperature of these small babies is maintained using incubators (Fig. 10.14) or initially with overhead radiant heaters. Incubators also allow ambient humidity to be provided, which reduces transepidermal heat loss.

Patent ductus arteriosus

The ductus arteriosus remains patent in many preterm infants. Shunting of blood across the ductus, from the left to the right side of the circulation, is most common in infants with RDS. It may produce no symptoms or it may cause apnoea and bradycardia, increased oxygen requirement and difficulty in weaning the infant from artificial ventilation. The pulses are 'bounding' from an increased pulse pressure, the precordial impulse becomes prominent and a systolic murmur may be audible. With increasing circulatory overload, signs of heart failure may develop. More accurate assessment of the infant's circulation can be obtained on echocardiography. If the infant is symptomatic, pharmacological closure with a prostaglandin synthetase inhibitor, indometacin or ibuprofen, is used. If these measures fail to close a symptomatic duct, surgical ligation will be required.

Fluid balance

A preterm infant's fluid requirements will vary with gestational and chronological age. On the first day of life, about 60–90 ml/kg is usually required. This is adjusted according to the infant's clinical condition, plasma electrolytes, urine output and weight change. Subsequent

Temperature control

Prevention of heat loss in newborn infants

1. ***Convection***
- Raise temperature of ambient air in incubator
- Clothe, including covering head
- Avoid draughts

2. ***Radiation***
- Cover baby
- Double walls for incubators

3. ***Evaporation***
- Dry and wrap at birth; if extremely preterm place baby's body directly into plastic bag at birth
- Humidify incubator

4. ***Conduction***
- Nurse on heated mattress.

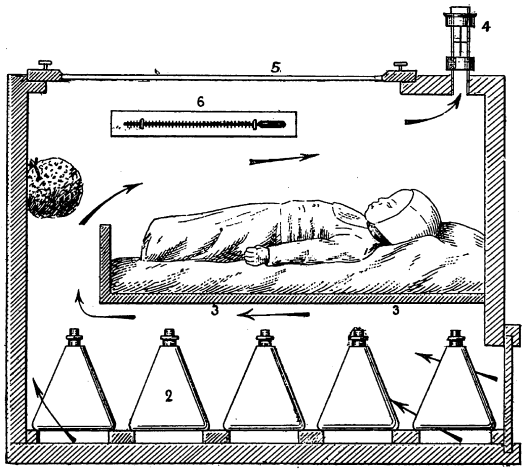


Figure 10.14 The importance of avoiding hypothermia in newborn infants has long been recognised. This incubator was used in the late nineteenth century to keep newborn infants warm. The sponge is to increase ambient humidity.

fluid volume is increased by 20–30 ml/kg per day to 150–180 ml/kg per day.

Nutrition

Preterm infants have a high nutritional requirement because of their rapid growth. Preterm infants at 28 weeks' gestation double their birthweight in 6 weeks and treble it in 12 weeks, whereas term babies double their weight in only 4.5 months and treble it in a year.

Infants of 35–36 weeks' gestational age are mature enough to suck and swallow milk. Less mature infants will need to be fed via an oro- or nasogastric tube. Even in very preterm infants, enteral feeds, preferably breast milk, are introduced as soon as possible. In these infants, breast milk needs to be supplemented with phosphate and may need supplementation with protein and calories (in breast milk fortifier) and calcium. In some neonatal units, extremely preterm infants are initially fed on donor breast milk if maternal breast milk is not available. If formula feeding is required, special infant formulas are available which are designed to meet the increased nutritional requirements of preterm infants but, in contrast to breast milk, do not provide protection against infection or other benefits of breast milk. In the very immature or sick infant, parenteral nutrition is often required. This is usually given through a central venous catheter, inserted peripherally (PICC lines, peripherally inserted central catheters), paying strict attention to aseptic technique both during insertion and when fluids are changed. However, PICC lines carry a significant risk of septicaemia; other risks include thrombosis of a major vein. For this reason, parenteral nutrition may sometimes be given via a peripheral vein, but extravasation may cause skin damage with scarring. Because of the significant risk of septicaemia from parenteral nutrition and the increased risk of necrotising enterocolitis with cow's milk based formula, mothers should be encouraged and supported to provide breast milk.

Poor bone mineralisation (osteopenia of prematurity) was previously common but is prevented by provision of adequate phosphate, calcium and vitamin D. Because iron is mostly transferred to the fetus during the last trimester, preterm babies have low iron stores and are at a risk of iron deficiency. This is in addition to loss of blood from sampling and an inadequate erythropoietin response. Iron supplements are started at several weeks of age and continued after discharge home.

Infection

Preterm infants are at increased risk of infection, as IgG is mostly transferred across the placenta in the last trimester and no IgA or IgM is transferred. In addition, infection in or around the cervix is often a reason for preterm labour and may cause infection shortly after birth. Most infections in preterm infants occur after several days of age and are nosocomial (hospital-derived); they are often associated with indwelling catheters or artificial ventilation. Infection is considered in more detail later in this chapter.

Infection in preterm infants is a major cause of death and contributes to bronchopulmonary dysplasia (chronic lung disease), white matter injury in the brain and later disability.

Preterm brain injury

Haemorrhages in the brain occur in 25% of very low birthweight infants and are easily recognised on cranial ultrasound scans (Fig. 10.15a). Typically, they occur in the germinal matrix above the caudate nucleus, which contains a fragile network of blood vessels. Most haemorrhages occur within the first 72 h of life. They are more common following perinatal asphyxia and in infants with severe respiratory distress syndrome. Pneumothorax is a significant risk factor. Small haemorrhages confined to the germinal matrix do not increase the risk of cerebral palsy. Haemorrhage may occur in the ventricles. The most severe haemorrhage is unilateral haemorrhagic infarction involving the parenchyma of the brain; this usually results in hemiplegia (Fig. 10.15b).

A large intraventricular haemorrhage may impair the drainage and reabsorption of cerebrospinal fluid (CSF), thus allowing CSF to build up under pressure. This dilatation (Fig. 10.15c) may resolve spontaneously or progress to hydrocephalus, which may cause the cranial sutures to separate, the head circumference to increase rapidly and the anterior fontanelle to become tense. A ventriculoperitoneal shunt may be required, but initially symptomatic relief may be provided by removal of CSF by lumbar puncture or ventricular tap. About half of infants with progressive post-haemorrhagic ventricular dilatation have cerebral palsy, a higher proportion if parenchymal infarction is also present.

Periventricular white matter brain injury may occur following ischaemia or inflammation and may occur in the absence of haemorrhage. It is more difficult to detect by cranial ultrasound. Initially there may be an echodense area or 'flare' within the brain parenchyma. This may resolve within a week (in which case the risk of cerebral palsy is not increased), but if cystic lesions become visible on ultrasound 2–4 weeks later, there is definite loss of white matter. Bilateral multiple cysts, called periventricular leukomalacia (PVL), have an 80–90% risk of spastic diplegia, often with cognitive impairment, if posteriorly sited (Fig. 10.15d).

Both intraventricular haemorrhage and periventricular leukomalacia may occur in the absence of abnormal clinical signs.

Necrotising enterocolitis

Necrotising enterocolitis is a serious illness mainly affecting preterm infants in the first few weeks of life. It is associated with bacterial invasion of ischaemic bowel wall. Preterm infants fed cow's milk formula are more likely to develop this condition than if they are fed only breast milk. The infant stops tolerating feeds, milk is aspirated from the stomach and there may be vomiting, which may be bile-stained. The abdomen becomes distended (Fig. 10.16a) and the stool

Cranial ultrasound in preterm infants

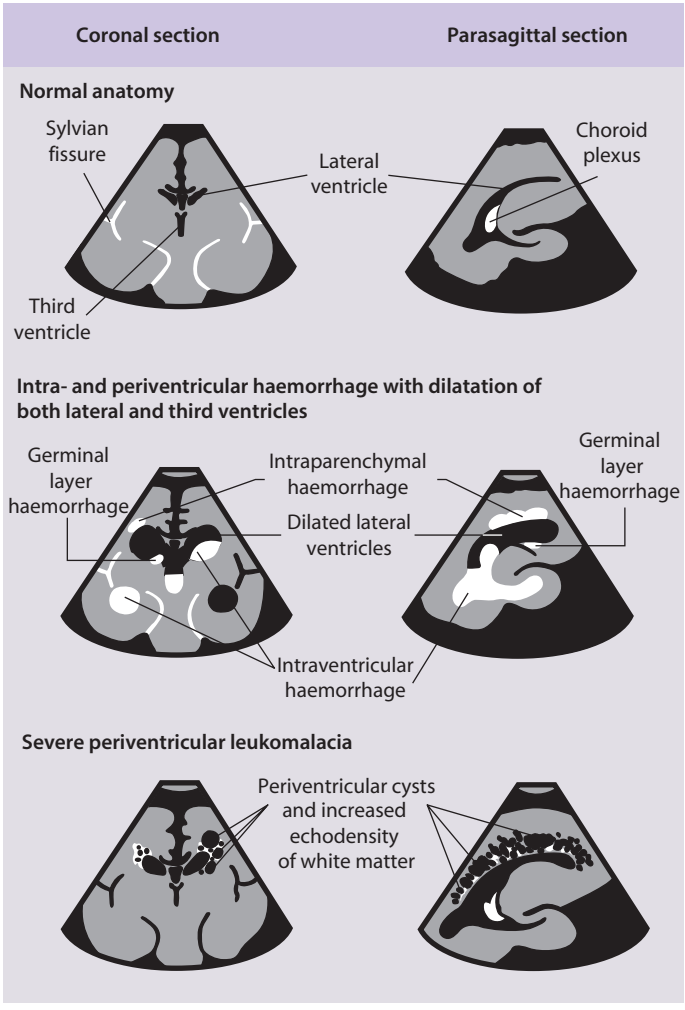


Figure 10.15a Cranial ultrasound in preterm infants.

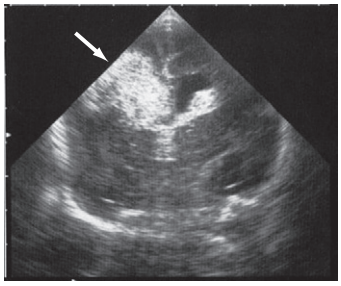


Figure 10.15b Large intraventricular haemorrhage with parenchymal haemorrhagic infarction on the right.

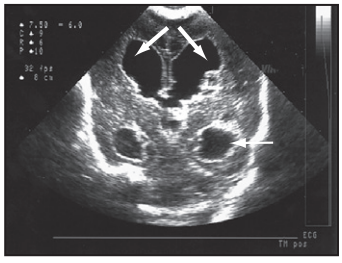


Figure 10.15c Dilatation of lateral ventricles following intraventricular haemorrhage.

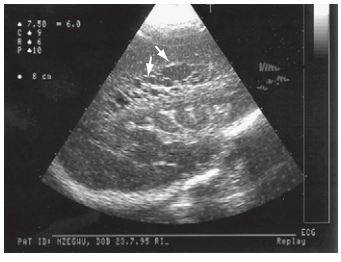


Figure 10.15d Widespread cysts in periventricular leukomalacia.

sometimes contains fresh blood. The infant may rapidly become shocked and require artificial ventilation because of abdominal distension and pain. The characteristic X-ray features are distended loops of bowel and thickening of the bowel wall with intramural gas, and there may be gas in the portal tract (Fig. 10.16b). The disease may progress to bowel perforation, which can be detected by X-ray or by transillumination of the abdomen.

Treatment is to stop oral feeding and give broad-spectrum antibiotics to cover both aerobic and anaerobic organisms. Parenteral nutrition is always needed and artificial ventilation and circulatory support are often needed. Surgery is performed for bowel perforation. The disease has significant morbidity and a mortality of about 20%. Long-term sequelae include the development of strictures and malabsorption if extensive bowel resection has been necessary.

Retinopathy of prematurity

Retinopathy of prematurity (ROP) affects developing blood vessels at the junction of the vascular and non-vascularised retina. There is vascular proliferation

Necrotising enterocolitis

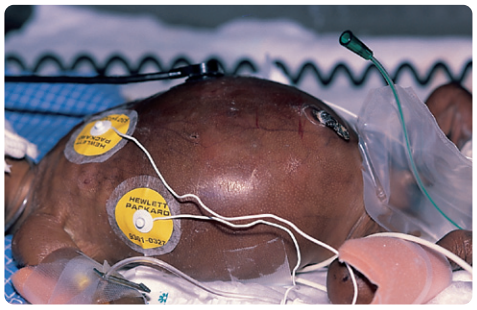


Figure 10.16a Necrotising enterocolitis showing gross abdominal distension and tense and shiny skin over the abdomen.

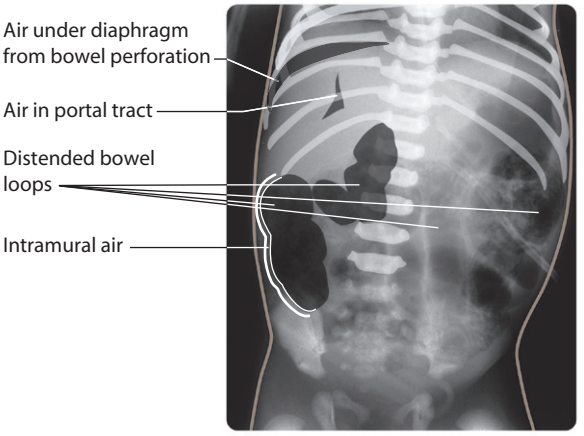


Figure 10.16b Diagram of characteristic features of necrotising enterocolitis on abdominal X-ray.

which may progress to retinal detachment, fibrosis and blindness. It was initially recognised that the risk is increased by uncontrolled use of high concentrations of oxygen. Now, even with careful monitoring of the infant's oxygenation, retinopathy of prematurity is still found in about 35% of all very low birthweight infants. As laser therapy reduces visual impairment, the eyes of susceptible preterm infants (<1500 g birthweight or <32 weeks' gestation) are screened every week by an ophthalmologist. Severe bilateral visual impairment occurs in about 1% of very low birthweight infants, mostly in infants of <28 weeks' gestation.

Bronchopulmonary dysplasia

Infants who still have an oxygen requirement at a post-menstrual age of 36 weeks are described as having bronchopulmonary dysplasia (BPD) or chronic lung disease. The lung damage comes from pressure and volume trauma from artificial ventilation, oxygen toxicity and infection. The chest X-ray characteristically shows widespread areas of opacification, sometimes with cystic changes (Fig. 10.17). Some infants need prolonged artificial ventilation, but most are weaned onto continuous positive airways pressure (CPAP) followed by additional ambient oxygen, sometimes over several months. Corticosteroid therapy may facilitate earlier weaning from the ventilator and often reduces the infant's oxygen requirements in the short term, but concern about increased risk of abnormal neurodevelopment including cerebral palsy limits use to those at highest risk and only short courses are given. Some babies go home while still receiving additional oxygen. A few infants with severe disease may die of intercurrent infection or pulmonary hypertension. Subsequent pertussis and RSV (respiratory syncytial virus) infection may cause respiratory failure necessitating intensive care.

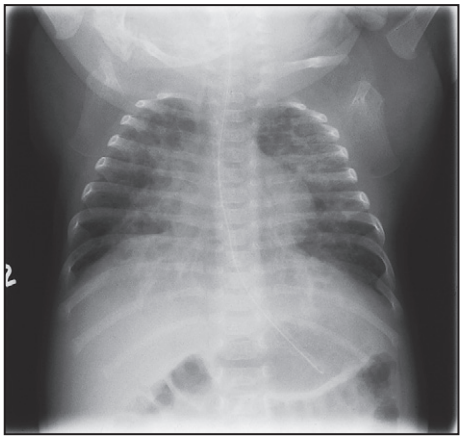


Figure 10.17 Chest X-ray of bronchopulmonary dysplasia (BPD) showing fibrosis and lung collapse, cystic changes and overdistension of the lungs.

Problems following discharge

To prevent anaemia of prematurity, additional iron as supplementation or in preterm formula is given until 6 months corrected age, when iron becomes available from solid foods. Multivitamins are also recommended. The standard immunisations should be given.

Medical problems include increased risk of:

- poor growth – at discharge, over 90% of VLBW infants are below the 10th centile for weight, length and head circumference. Many show some catch-up growth in the first 2–3 years
- pneumonia/wheezing/asthma
- bronchiolitis from RSV (respiratory syncytial virus) infection (hospitalisation reduced

by giving RSV monoclonal antibody, palivizumab)
- bronchopulmonary dysplasia – may require additional oxygen therapy for many months, when it may be provided at home
- gastro-oesophageal reflux – especially with bronchopulmonary dysplasia
- complex nutritional and gastrointestinal disorders – following necrotising enterocolitis or gastrointestinal surgery
- inguinal hernias – require surgical repair.

Readmission to hospital during the first year of life is increased approximately four-fold – mainly for respiratory disorders and surgical repair of inguinal hernias.

Summary

Summary of problems of very low birthweight infants (<1.5 kg)

Respiratory

Respiratory distress syndrome (surfactant deficiency)(74%)
- respiratory distress within 4 hours of birth
- antenatal corticosteroids and surfactant therapy reduce morbidity and mortality
- oxygen therapy, but excess may damage the retina
- nasal CPAP (continuous positive airway pressure) (67%) and mechanical ventilation (64%) - often required to expand lungs and prevent lung collapse

Pneumothorax (4%)
Apnoea and bradycardia and desaturations
Bronchopulmonary dysplasia (chronic lung disease) – O_2 requirement at 36 weeks post-menstrual age (27%)

Circulation

Hypotension – may require volume support, intropes or corticosteroids
Patent ductus arteriosus – needing medical treatment (34%) or surgical ligation (8%)

Nutrition

Nasogastric tube feeding – until 35–36 weeks post-menstrual age
Feeding intolerance - TPN (total parenteral nutrition) often required

Gastrointestinal

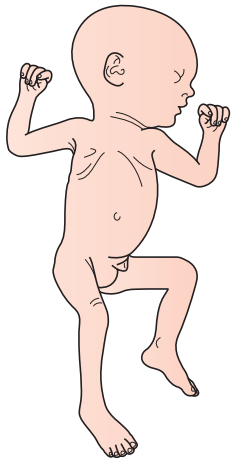
Necrotising enterocolitis (6%)
– serious, management is medical or surgery for bowel necrosis or perforation

Metabolic

Hypoglycaemia – common
Electrolyte disturbances
Osteopenia of prematurity from phosphate deficiency

Hearing

Checked before discharge



Jaundice – common, low treatment threshold

Anaemia
Often need blood transfusions

Eyes

Retinopathy of prematurity – may need laser therapy (4%)

Temperature control

Avoid hypothermia
Nurse in neutral thermal environment
Nurse in incubator or under radiant warmer
Clothe if possible
Humidity reduces evaporative heat loss

Infection

Common and potentially serious (25%)
Increased risk of early-onset infection – group B streptococcus
Main problem is nosocomial infection – mainly coagulase-negative staphylococcus, also fungal and other infections

Brain injury

Haemorrhage (25%)
- germinal layer, intraventricular, parenchymal

Ventricular dilatation – may need ventriculo-peritoneal shunt
Periventricular leukomalacia (3%)
– ischaemic white matter injury

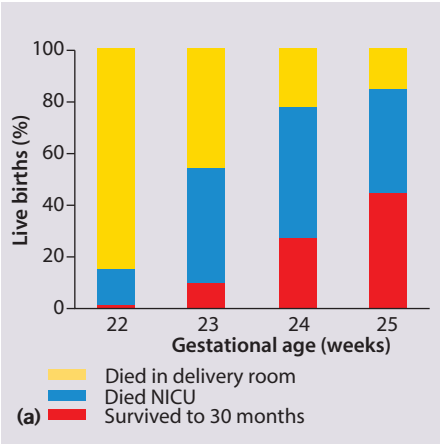
Following discharge

Specialist community nursing support helpful, if available
Increased risk of respiratory infection and wheezing – especially from bronchiolitis (caused by respiratory syncitial virus, RSV) and pertussis; may need intensive care
Consider prophylaxis against RSV infection
Increased rehospitalisation – respiratory disorders, inguinal hernias
Monitor growth, development (for learning disorders, co-ordination, cerebral palsy), behaviour, attention, vision, hearing – increased risk of impairment

About 5–10% of very low birthweight infants develop cerebral palsy, but the most common impairments are learning difficulties. The prevalence of cognitive impairment and of other associated difficulties increases with decreasing gestational age at birth, and is greatest if born at very early gestational age (<26 weeks' gestation) (Fig. 10.18a,b). It becomes increasingly evident when the individual child is compared to their peers at nursery or school. In addition, children may have difficulties with:

- fine motor skills, e.g. threading beads
- concentration, with short attention span
- behavior problems, especially attention deficit disorders
- abstract reasoning, e.g. mathematics
- processing several tasks simultaneously.

A small proportion also have hearing impairment, with 1–2% requiring amplification, or visual impairment, with 1% blind in both eyes. A greater proportion have refraction errors and squints and therefore require glasses.



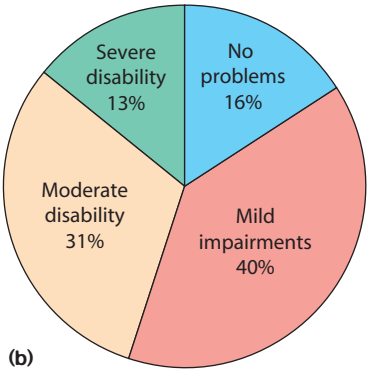


Figure 10.18 The EPICure study, a population-based study of mortality and disability in the UK and Ireland in 1995 of all infants born alive at 22–25 weeks' gestation. **(a)** Survival. (Adapted from Wood NS, et al. 2000. Neurologic and developmental disability after extremely preterm birth. EPICure Study Group. *New England Journal of Medicine* 343:378–384). **(b)** Proportion of survivors with disability at 11 years of age. (Adapted from Johnson S et al. 2009. Neurodevelopmental disability through 11 years of age in children born before 26 weeks of gestation. *Pediatrics* 124:e249–e257). Preterm babies born in 2006 show a 30% increase in numbers born at <26 weeks and increased survival. Long-term follow-up is in progress.

Jaundice

Over 50% of all newborn infants become visibly jaundiced. This is because:

- there is marked physiological release of haemoglobin from the breakdown of red cells because of the high Hb concentration at birth (Fig. 10.19)
- the red cell life span of newborn infants (70 days) is markedly shorter than that of adults (120 days)
- hepatic bilirubin metabolism is less efficient in the first few days of life.

Neonatal jaundice is important as:

- it may be a sign of another disorder, e.g. haemolytic anaemia, infection, metabolic disease, liver disease.
- unconjugated bilirubin can be deposited in the brain, particularly in the basal ganglia, causing kernicterus.

Kernicterus

This is the encephalopathy resulting from the deposition of unconjugated bilirubin in the basal ganglia and brainstem nuclei (Fig. 10.20). It may occur when the level of unconjugated bilirubin exceeds the albumin-binding capacity of bilirubin of the blood. As this free bilirubin is fat-soluble, it can cross the blood–brain barrier. The neurotoxic effects vary in severity from transient disturbance to severe damage and death. Acute manifestations are lethargy and poor feeding. In severe cases, there is irritability, increased muscle tone causing the baby to lie with an arched back (opisthotonos), seizures and coma. Infants who survive may develop choreoathetoid cerebral palsy (due to damage to the basal ganglia), learning difficulties and sensorineural deafness. Kernicterus used to be an important cause of brain damage in infants with severe rhesus haemolytic disease, but has become rare since the introduction of prophylactic anti-D immunoglobulin for rhesus-negative mothers. However, a few cases continue to occur, especially in slightly preterm infants (35–37 weeks), which has led NICE to issue guidelines on the management of neonatal jaundice.

Clinical evaluation

Babies become clinically jaundiced when the bilirubin level reaches about 80 µmol/L. Management varies according to the infant's gestational age, age at onset, bilirubin level and rate of rise, and the overall clinical condition.

Age at onset

The age of onset is a useful guide to the likely cause of the jaundice (Table 10.2).

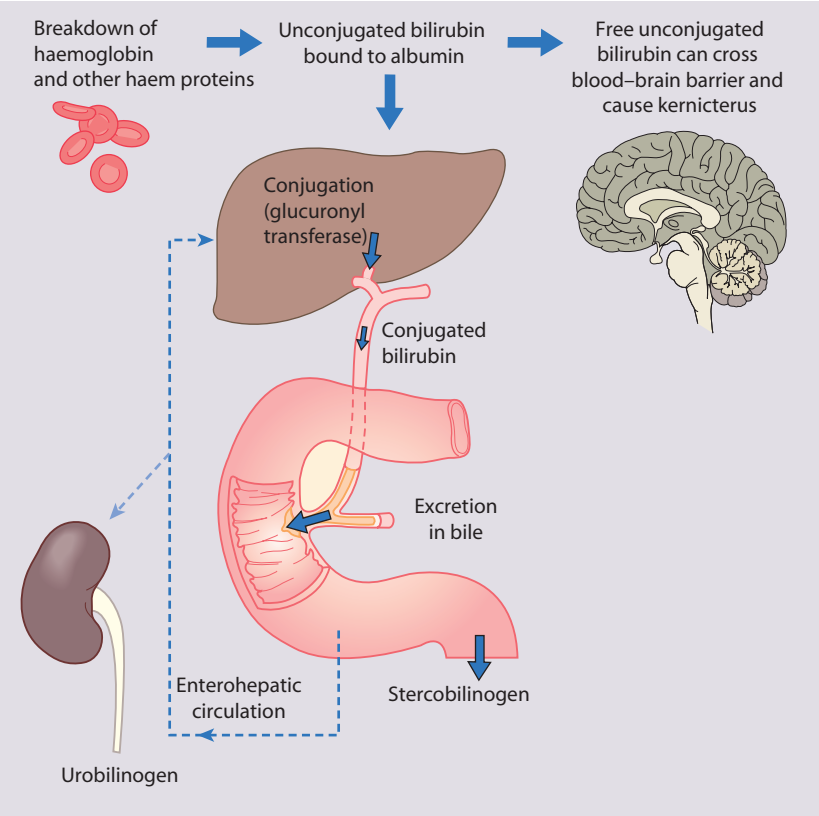


Figure 10.19 The breakdown product of haemoglobin is unconjugated bilirubin (indirect bilirubin), which is insoluble in water but soluble in lipids. It is carried in the blood bound to albumin. When the albumin binding is saturated, free unconjugated bilirubin can cross the blood–brain barrier, as it is lipid soluble. Unconjugated bilirubin bound to albumin is taken up by the liver and conjugated by the enzyme glucuronyl transferase to conjugated bilirubin (direct bilirubin), which is water-soluble and excreted in bile into the gut and is detectable in urine when blood levels rise. Reabsorption of bilirubin from the gut (enterohepatic circulation) is increased when milk intake is low.

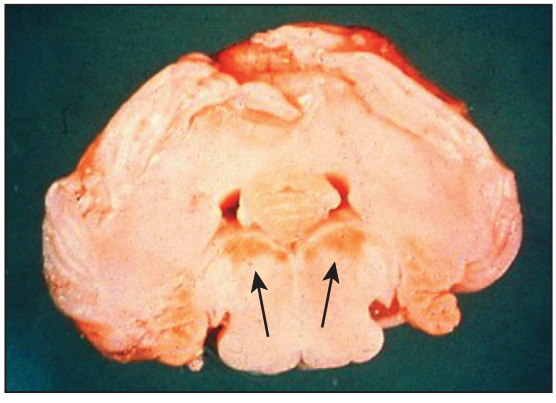


Figure 10.20 Postmortem brainstem and cerebellum showing kernicterus with yellow bilirubin staining of brainstem nuclei (arrows).

Jaundice <24 h of age

Jaundice starting within 24 h of birth usually results from haemolysis. This is particularly important to identify as the bilirubin is unconjugated and can rise very rapidly and reach extremely high levels.

Haemolytic disorders

Rhesus haemolytic disease – Affected infants are usually identified antenatally and monitored and treated if necessary (see Ch. 9). The birth of a severely affected infant, with anaemia, hydrops and hepatosplenomegaly with rapidly developing severe jaundice, has become rare. Antibodies may develop to rhesus antigens other than D and to the Kell and Duffy blood groups, but haemolysis is usually less severe.

ABO incompatibility – This is now more common than rhesus haemolytic disease. Most ABO antibodies are IgM and do not cross the placenta, but some group O women have an IgG anti-A-haemolysin in the blood which can cross the placenta and haemolyse the red cells of a group A infant. Occasionally, group B infants are affected by anti-B haemolysins. Haemolysis can cause severe jaundice but it is usually less severe than in rhesus disease. The infant's haemoglobin level is usually normal or only slightly reduced and, in contrast to rhesus disease, hepatosplenomegaly is absent. The direct antibody test (Coombs' test), which demonstrates antibody on the surface of red cells, is positive. The jaundice usually peaks in the first 12–72 h.

G6PD deficiency (see Ch. 22) – Mainly in people originating in the Mediterranean, Middle-East and Far East

Table 10.2 Causes of neonatal jaundice

Jaundice starting at <24 h of age	*Haemolytic disorders*:
	Rhesus incompatibility
	ABO incompatibility
	G6PD deficiency
	Spherocytosis, pyruvate kinase deficiency
	Congenital infection
Jaundice at 24 h to 2 weeks of age	Physiological jaundice
	Breast milk jaundice
	Infection, e.g. urinary tract infection
	Haemolysis, e.g. G6PD deficiency, ABO incompatibility
	Bruising
	Polycythaemia
	Crigler–Najjar syndrome
Jaundice at >2 weeks of age	*Unconjugated:*
	Physiological or breast milk jaundice
	Infection (particularly urinary tract)
	Hypothyroidism
	Haemolytic anaemia, e.g. G6PD deficiency
	High gastrointestinal obstruction, e.g. pyloric stenosis
	Conjugated (>25 μmol/L):
	Bile duct obstruction
	Neonatal hepatitis

or in African-Americans. Mainly affects male infants , but some females develop significant jaundice. Parents of affected infants should be given a list of drugs to be avoided, as they may precipitate haemolysis.

Spherocytosis – This is considerably less common than G6PD deficiency (see Ch. 22). There is often, but not always, a family history. The disorder can be identified by recognising spherocytes on the blood film.

Congenital infection

Jaundice at birth can also be from congenital infection. In this case, the bilirubin is conjugated and the infants have other abnormal clinical signs, such as growth restriction, hepatosplenomegaly and thrombocytopenic purpura.

Jaundice at 2 days to 2 weeks of age

Physiological jaundice

Most babies who become mildly or moderately jaundiced during this period have no underlying cause and the bilirubin has risen as the infant is adapting to the transition from fetal life. The term 'physiological jaundice' can only be used after other causes have been considered.

Breast milk jaundice

Jaundice is more common and more prolonged in breast-fed infants. The hyperbilirubinaemia is unconjugated. The cause is multifactorial but may involve increased enterohepatic circulation of bilirubin.

Dehydration

In some infants, the jaundice is exacerbated if milk intake is poor from a delay in establishing breast-feeding and the infant becomes dehydrated. Breast-feeding should be continued, although the bilirubin level would fall if it were interrupted. In some infants, intravenous fluids are needed to correct dehydration.

Infection

An infected baby may develop an unconjugated hyperbilirubinaemia from poor fluid intake, haemolysis, reduced hepatic function and an increase in the enterohepatic circulation. If infection is suspected, appropriate investigations and treatment should be instigated. In particular, urinary tract infection may present in this way.

Other causes

Although jaundice from haemolysis usually presents in the first day of life, it may occur during the first week. Bruising and polycythaemia (venous haematocrit is >0.65) will exacerbate the infant's jaundice. The very rare Crigler–Najjar syndrome, in which the enzyme glucuronyl transferase is deficient or absent, may result in extremely high levels of unconjugated bilirubin.

The causes and management of jaundice at >2 weeks of age (persistent neonatal jaundice), (3 weeks if preterm), are different and are considered separately below.

Severity of jaundice

Jaundice can be observed most easily by blanching the skin with one's finger. The jaundice tends to start on the head and face and then spreads down the trunk and limbs. If the baby is clinically jaundiced, the bilirubin should be checked with a transcutaneous bilirubin meter or blood sample. It is easy to underestimate in Afro-Caribbean, Asian and preterm babies, and a low threshold should be adopted for measuring the bilirubin of these infants. A high transcutaneous bilirubin level must be checked with a blood laboratory measurement. It is now recommended in the UK that all babies should be checked clinically for jaundice in the first 72 h of life, whether at hospital or home, and if clinically jaundiced, a transcutaneous measurement made.

Rate of change

The rate of rise tends to be linear until a plateau is reached, so serial measurements can be plotted on a chart and used to anticipate the need for treatment before it rises to a dangerous level.

Gestation

Preterm infants are more susceptible to damage from raised bilirubin, so the intervention threshold is lower.

Clinical condition

Infants who experience severe hypoxia, hypothermia or any serious illness may be more susceptible to damage from severe jaundice. Drugs which may displace bilirubin from albumin, e.g. sulphonamides and diazepam, are therefore avoided in newborn infants.

Management

Poor milk intake and dehydration will exacerbate jaundice and should be corrected, but studies have failed to show that routinely supplementing breast-fed infants with water or dextrose solution reduces jaundice. Phototherapy is the most widely used therapy, with exchange transfusion for severe cases.

Phototherapy

Light (wavelength 450 nm) from the blue–green band of the visible spectrum converts unconjugated bilirubin into a harmless water-soluble pigment excreted predominantly in the urine. It is delivered with an overhead light source placed the optimal distance above the infant to achieve high irradiance. Although no long-term sequelae of phototherapy from overhead light have been reported, it is disruptive to normal nursing of the infant and should not be used indiscriminately. The infant's eyes are covered, as bright light is uncomfortable. Phototherapy can result in temperature instability as the infant is undressed, a macular rash and bronze discoloration of the skin if the jaundice is conjugated.

Continuous multiple ('intensive') phototherapy is given if the bilirubin is rising rapidly or has reached a high level.

Exchange transfusion

Exchange transfusion is required if the bilirubin rises to levels which are considered potentially dangerous. Blood is removed from the baby in small aliquots, (usually from an arterial line or the umbilical vein) and replaced with donor blood (via peripheral or umbilical vein). Twice the infant's blood volume (2×80 ml/kg) is exchanged. Donor blood should be as fresh as possible and screened to exclude CMV, hepatitis B and C and HIV infection. The procedure does carry some risk of morbidity and mortality.

Phototherapy has been very successful in reducing the need for exchange transfusion. In infants with rhesus haemolytic disease or ABO incompatibility unresponsive to intensive phototherapy, intravenous immunoglobulin reduces the need for exchange transfusion.

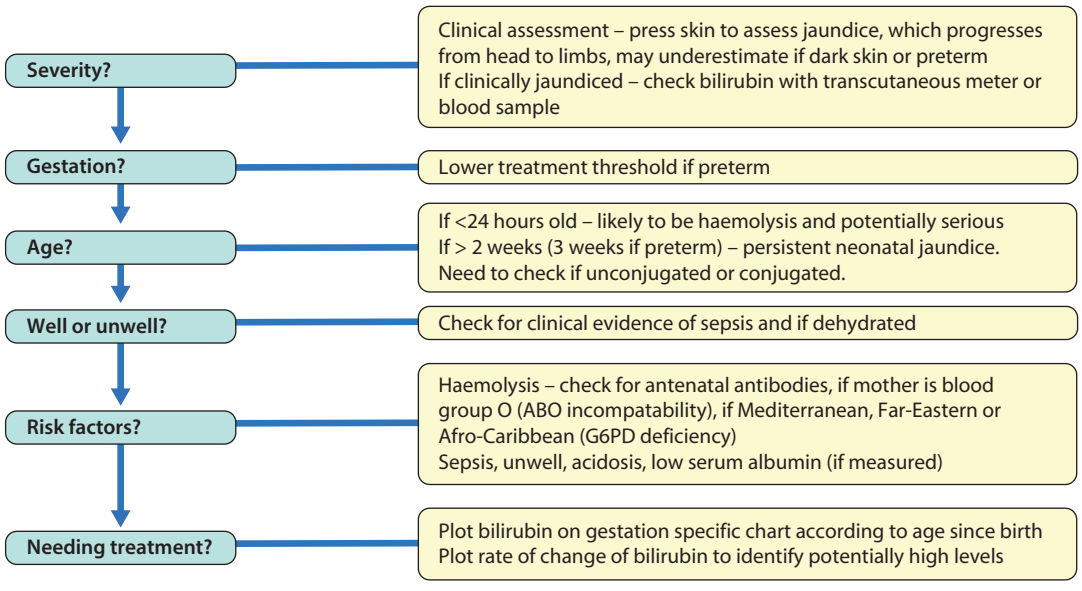
There is no bilirubin level known to be safe or which will definitely cause kernicterus. In rhesus haemolytic disease, it was found that kernicterus could be prevented if the bilirubin was kept below 340 µmol/L (20 mg/dl). As there is no consensus among paediatricians in the UK on the bilirubin levels for phototherapy and exchange transfusion, guidelines have been published by NICE to ensure uniform practice.

Jaundice at >2 weeks of age

Jaundice in babies more than 2 weeks old (3 weeks if preterm), is called persistent or prolonged neonatal jaundice. The key feature is that it may be caused by biliary atresia, and it is important to diagnose

Summary

Assessment of neonatal jaundice



biliary atresia promptly, as delay in surgical treatment adversely affects outcome (see Ch. 20 for further details).

However, in most infants with persistent neonatal jaundice, the hyperbilirubinaemia is unconjugated, but this needs to be confirmed on laboratory testing.

In prolonged unconjugated hyperbilirubinaemia:

- 'Breast milk jaundice' is the most common cause, affecting up to 15% of healthy breast-fed infants; the jaundice gradually fades and disappears by 4–5 weeks of age.
- Infection, particularly of the urinary tract, needs to be considered.
- Congenital hypothyroidism may cause prolonged jaundice before the clinical features of coarse facies, dry skin, hypotonia and constipation become evident. Affected infants should be identified on routine neonatal biochemical screening (Guthrie test).

Conjugated hyperbilirubinaemia (>25 µmol/L) is suggested by the baby passing dark urine and unpigmented pale stools. Hepatomegaly and poor weight gain are other clinical signs that may be present. Its causes include neonatal hepatitis syndrome and biliary atresia, with improved prognosis of biliary atresia with early diagnosis (see Ch. 20 for further details).

Respiratory distress in term infants

Newborn infants with respiratory problems develop the following signs of respiratory distress:

- tachypnoea (>60 breaths/min)
- laboured breathing, with chest wall recession (particularly sternal and subcostal indrawing) and nasal flaring
- expiratory grunting
- cyanosis if severe.

The causes in term infants are listed in Table 10.3.

Affected infants should be admitted to the neonatal unit for monitoring of heart and respiratory rates, oxygenation and circulation. A chest X-ray will be required to help identify the cause, especially those causes which may need immediate treatment, e.g. pneumothorax or diaphragmatic hernia. Additional ambient oxygen, mechanical ventilation and circulatory support are given as required.

Transient tachypnoea of the newborn

This is by far the commonest cause of respiratory distress in term infants. It is caused by delay in the resorption of lung liquid and is more common after birth by Caesarean section. The chest X-ray may show fluid in the horizontal fissure. Additional ambient oxygen may be required. The condition usually settles within the first day of life but can take several days to resolve completely. This is a diagnosis made after consideration and exclusion of other causes.

Table 10.3 Causes of respiratory distress in term infants

Pulmonary	
Common	Transient tachypnoea of the newborn
Less common	Meconium aspiration
	Pneumonia
	Respiratory distress syndrome
	Pneumothorax
	Persistent pulmonary hypertension of the newborn
	Milk aspiration
Rare	Diaphragmatic hernia
	Tracheo-oesophageal fistula (TOF)
	Pulmonary hypoplasia
	Airways obstruction, e.g. choanal atresia
	Pulmonary haemorrhage
Non-pulmonary	
	Congenital heart disease
	Intracranial birth trauma/ encephalopathy
	Severe anaemia
	Metabolic acidosis

Meconium aspiration

Meconium is passed before birth by 8–20% of babies. It is rarely passed by preterm infants, and occurs increasingly the greater the gestational age, affecting 20–25% of deliveries by 42 weeks. It may be passed in response to fetal hypoxia. At birth these infants may inhale thick meconium. Asphyxiated infants may start gasping and aspirate meconium before delivery. Meconium is a lung irritant and results in both mechanical obstruction and a chemical pneumonitis, as well as predisposing to infection. In meconium aspiration the lungs are over-inflated, accompanied by patches of collapse and consolidation. There is a high incidence of air leak, leading to pneumothorax and pneumomediastinum. Artificial ventilation is often required. Infants with meconium aspiration may develop persistent pulmonary hypertension of the newborn which may make it difficult to achieve adequate oxygenation despite high pressure ventilation (see below for management). Severe meconium aspiration is associated with significant morbidity and mortality.

Pneumonia

Prolonged rupture of the membranes, chorioamnionitis and low birthweight predispose to pneumonia. Infants with respiratory distress will usually require

investigation to identify any infection. Broad-spectrum antibiotics are started early until the results of the infection screen are available.

Pneumothorax

A pneumothorax may occur spontaneously in up to 2% of deliveries. It is usually asymptomatic but may cause respiratory distress. Pneumothoraces also occur secondary to meconium aspiration, respiratory distress syndrome or as a complication of ventilation. (Management is described earlier in this chapter.)

Milk aspiration

This occurs more frequently in preterm infants and those with respiratory distress or neurological damage. Babies with bronchopulmonary dysplasia often have gastro-oesophageal reflux, which predisposes to aspiration. Infants with a cleft palate are prone to aspirate respiratory secretions or milk.

Persistent pulmonary hypertension of the newborn

This life-threatening condition is usually associated with birth asphyxia, meconium aspiration, septicaemia or respiratory distress syndrome. It sometimes occurs as a primary disorder. As a result of the high pulmonary vascular resistance, there is right-to-left shunting within the lungs and at atrial and ductal levels. Cyanosis occurs soon after birth. Heart murmurs and signs of heart failure are often absent. A chest X-ray shows that the heart is of normal size and there may be pulmonary oligaemia. An urgent echocardiogram is required to establish that the child does not have congenital heart disease.

Most infants require mechanical ventilation and circulatory support in order to achieve adequate oxygenation. Inhaled nitric oxide, a potent vasodilator, is often beneficial. Another vasodilator, sildenafil (Viagra), has been introduced more recently. High-frequency or oscillatory ventilation is sometimes helpful. Extracorporeal membrane oxygenation (ECMO), where the infant is placed on heart and lung bypass for several days, is indicated for severe but reversible cases, but is only performed in a few specialist centres.

Diaphragmatic hernia

This occurs in about 1 in 4000 births. Many are now diagnosed on antenatal ultrasound screening. In the newborn period, it usually presents with failure to respond to resuscitation or as respiratory distress. In most cases, there is a left-sided herniation of abdominal contents through the posterolateral foramen of the diaphragm. The apex beat and heart sounds will then be displaced to the right side of the chest, with poor air entry in the left chest. Vigorous resuscitation may cause a pneumothorax in the normal lung, thereby aggravating the situation. The diagnosis is confirmed by X-ray of the chest and abdomen (Fig. 10.21). Once the diagnosis is suspected, a large nasogastric tube is passed and suction is applied to prevent distension of the intrathoracic bowel. After stabilisation, the diaphragmatic hernia is repaired surgically, but in most infants with this condition the main problem is pulmonary hypoplasia – where compression by the herniated viscera throughout pregnancy has prevented development of the lung in the fetus. If the lungs are hypoplastic, mortality is high.

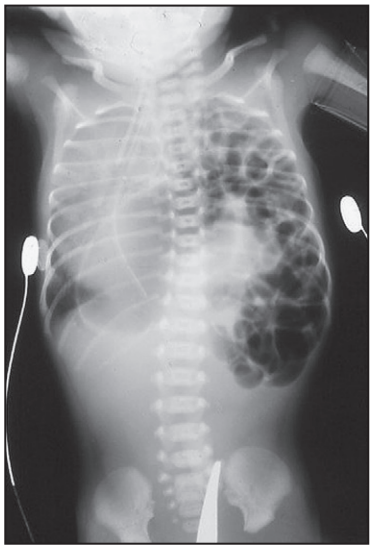


Figure 10.21 Chest X-ray of diaphragmatic hernia showing loops of bowel in the left chest and displacement of the mediastinum.

Other causes

Other causes of respiratory distress are listed in Table 10.3. When due to heart failure, abnormal heart sounds and/or heart murmurs may be present on auscultation. An enlarged liver from venous congestion is a helpful sign. The femoral arteries must be palpated in all infants with respiratory distress, as coarctation of the aorta and interrupted aortic arch are important causes of heart failure in newborn infants.

Infection

The time of highest risk in childhood for acquiring a serious invasive bacterial infection is the neonatal period. Infections fall into two broad categories, early- and late-onset sepsis.

Early-onset infection

In early-onset sepsis (<48 h after birth), bacteria have ascended from the birth canal and invaded the amniotic fluid. The fetus is secondarily infected because the fetal lungs are in direct contact with infected amniotic fluid. These infants have pneumonia and secondary bacteraemia/septicaemia. In contrast, congenital viral infections and early-onset infection with *Listeria*

Box 10.2 Clinical features of neonatal sepsis

- Fever or temperature instability or hypothermia
- Poor feeding
- Vomiting
- Apnoea and bradycardia
- Respiratory distress
- Abdominal distension
- Jaundice
- Neutropenia
- Hypo-/hyperglycaemia
- Shock
- Irritability
- Seizures
- Lethargy, drowsiness
- *In meningitis:*
- Tense or bulging fontanelle
- Head retraction (opisthotonos)

monocytogenes, fetal infection is acquired via the placenta following maternal infection.

The risk of early-onset infection is increased if there has been prolonged or premature rupture of the amniotic membranes, and when chorioamnionitis is clinically evident such as when the mother has fever during labour. Presentation is with respiratory distress, apnoea and temperature instability (Box 10.2). A chest X-ray is performed, together with a septic screen. A full blood count is performed to detect neutropenia, as well as blood cultures. An acute-phase reactant (C-reactive protein) is helpful but takes 12–24 h to rise, so one normal result does not exclude infection, but two consecutive normal values are strong evidence against infection. Antibiotics are started immediately without waiting for culture results. Intravenous antibiotics are given to cover group B streptococci, *Listeria monocytogenes* and other Gram-positive organisms (usually benzylpenicillin or amoxicillin), combined with cover for Gram-negative organisms (usually an aminoglycoside such as gentamicin). If cultures and CRP are negative and the infant has recovered clinically, antibiotics can be stopped after 48 h. If the blood culture is positive or if there are any neurological signs, CSF must be examined and cultured.

Late-onset infection

In late-onset infection (>48 h after birth), the source of infection is often the infant's environment. The presentation is usually non-specific (Box 10.2). Nosocomially acquired infections are an inherent risk in a neonatal unit, and all staff must adhere strictly to effective hand hygiene measures to prevent cross-infection. In neonatal intensive care, the main sources of infection are indwelling central venous catheters for parenteral nutrition, invasive procedures which break the protective barrier of the skin, and tracheal tubes. Coagulase-negative staphylococcus (*Staphylococcus epidermidis*) is the most common pathogen, but the range of organisms is broad, and includes Gram-positive bacteria (*Staphylococcus aureus* and *Enterococcus faecalis*) and Gram-negative bacteria (*Escherichia coli* and *Pseudomonas*, *Klebsiella* and *Serratia* species). Initial therapy (e.g. with flucloxacillin and gentamicin) is aimed to cover most staphylococci and Gram-negative bacilli. If the organism is resistant to these antibiotics or the infant's condition does not improve, specific antibiotics (e.g. vancomycin for coagulase-negative staphylococci or enterococci) or broad-spectrum antibiotics (e.g. meropenem) may be indicated. Use of prolonged or broad-spectrum antibiotics predisposes to invasive fungal infections (e.g. *Candida albicans*) in premature babies. Serial measurements of an acute-phase reactant (CRP) are useful to monitor response to therapy.

Neonatal meningitis, although uncommon, has a mortality of 20–50%, with one-third of survivors having serious sequelae. Presentation is non-specific (Box 10.2); a bulging fontanelle and hyperextension of neck and back (opisthotonos) are late signs and rarely seen in newborn infants. If meningitis is thought likely, ampicillin or penicillin and a third-generation cephalosporin (e.g. cefotaxime, which has CSF penetration) are given. Complications include cerebral abscess, ventriculitis, hydrocephalus, hearing loss and neurodevelopmental impairment.

Some specific infections

Group B streptococcal infection

Around 10–30% of pregnant women have faecal or vaginal carriage of group B streptococci. The organism causes early- and late-onset sepsis. In early-onset sepsis, the newborn baby has respiratory distress and pneumonia. In the UK, approximately 0.5–1 in 1000 babies have early-onset infection; most have pneumonia only, but it may cause septicaemia and meningitis. The severity of the neonatal presentation depends on the duration of the infection in utero. Mortality in babies with positive blood or CSF cultures is up to 10%.

Up to half of infants born to mothers who carry group B streptococcus are colonised on their mucous membranes or skin. Some of these babies develop late-onset disease, up to 3 months of age. It usually presents with meningitis, or occasionally with focal infection (e.g. osteomyelitis or septic arthritis).

In colonised mothers, risk factors for infection are preterm, prolonged rupture of membranes, maternal fever during labour (>38° C), maternal chorioamnionitis or previously infected infant. Prophylactic intrapartum antibiotics given intravenously to the mother can prevent group B streptococcus infection in the newborn baby. There are two approaches to the use of intrapartum antibiotics – universal screening at 35–38 weeks to identify mothers who carry the organism (as practiced in the USA and Australia) and a risk-based approach, in which mothers with risk factors for infection are offered antibiotics (as in the UK).

Listeria monocytogenes *infection*

Fetal or newborn *Listeria* infection is uncommon but serious. The organism is transmitted to the mother in food, such as unpasteurised milk, soft cheeses

and undercooked poultry. It causes a bacteraemia, often with mild, influenza-like illness in the mother, and passage to the fetus via the placenta. Maternal infection may cause spontaneous abortion, preterm delivery or fetal/neonatal sepsis. Characteristic features are meconium staining of the liquor, unusual in preterm infants, a widespread rash, septicaemia, pneumonia and meningitis. The mortality is 30%.

Late-onset disease also occurs, most often with meningitis, and has a better prognosis.

Gram-negative infections

Early-onset infection is acquired in the same way as group B streptococcal infection. Late-onset infection is usually from infected central venous lines, but occasionally from seeding to the circulation from the intestines.

Conjunctivitis

Sticky eyes are common in the neonatal period, starting on the 3rd or 4th day of life. Cleaning with saline or water is all that is required and the condition resolves spontaneously. A more troublesome discharge with redness of the eye may be due to staphylococcal or streptococcal infection and can be treated with a topical antibiotic eye ointment, e.g. neomycin.

Purulent discharge with conjunctival injection and swelling of the eyelids within the first 48 h of life may be due to gonococcal infection. The discharge should be Gram-stained urgently, as well as cultured, and treatment started immediately, as permanent loss of vision can occur. In countries such as the UK and the USA where penicillin resistance is a problem, a third-generation cephalosporin is given intravenously. The eye needs to be cleansed frequently.

Chlamydia trachomatis eye infection usually presents with a purulent discharge, together with swelling of the eyelids (Fig. 10.22), at 1–2 weeks of age, but may also present shortly after birth. The organism can be identified with immunofluorescent staining. Treatment is with oral erythromycin for 2 weeks. The mother and partner also need to be checked and treated.



Figure 10.22 Purulent discharge, together with swollen eyelids, in an 8-day-old infant. This is the characteristic presentation of conjunctivitis from *Chlamydia trachomatis*. *Neisseria gonorrhoeae* was absent.

Umbilical infection

The umbilicus dries and separates during the first few days of life. If the skin surrounding the umbilicus becomes inflamed, systemic antibiotics are indicated. Sometimes the umbilicus continues to be sticky, as it is prevented from involuting by an umbilical granuloma. This can be removed by applying silver nitrate while protecting the surrounding skin to avoid chemical burns, or by applying a ligature around the base of the exposed stump.

Herpes simplex virus (HSV) infections

Neonatal HSV infection is uncommon, occurring in 1 in 3000 to 1 in 20 000 live births. HSV infection is usually transmitted during passage through an infected birth canal or occasionally by ascending infection. The risk to an infant born to a mother with a primary genital infection is high, about 40%, while the risk from recurrent maternal infection is less than 3%. In most infants who develop HSV infection, the condition is unexpected as the mother does not know that she is infected (asymptomatic or non-specific illness).

Infection is more common in preterm infants. Presentation is at any time up to 4 weeks of age, with localised herpetic lesions on the skin or eye, or with encephalitis or disseminated disease. Mortality due to localised disease is low, but, even with aciclovir treatment, disseminated disease has a high mortality with considerable morbidity after encephalitis. If the mother is recognised as having primary disease or develops genital herpetic lesions at the time of delivery, elective Caesarean section is indicated. Women with a history of recurrent genital infection can be delivered vaginally as the risk of neonatal infection is low and maternal treatment before delivery minimises the presence of virus at delivery. Aciclovir can be given prophylactically to the baby during the at-risk period, but its efficacy is unproven.

Hepatitis B

Infants of mothers who are hepatitis B surface antigen (HBsAg)-positive should receive hepatitis B vaccination shortly after birth to prevent vertical transmission. The vaccination course needs to be completed during infancy and antibody response checked. Babies are at highest risk of becoming chronic carriers when their mothers are 'e' antigen-positive but have no 'e' antibodies. Infants of 'e' antigen-positive mothers should also be given passive immunisation with hepatitis B immunoglobulin within 24 h of birth.

Infants of HBsAg-positive mothers should be vaccinated against hepatitis B.

Hypoglycaemia

Hypoglycaemia is particularly likely in the first 24 h of life in babies with intrauterine growth restriction, who are preterm, born to mothers with diabetes mellitus,

are large-for-dates, hypothermic, polycythaemic or ill for any reason. Growth-restricted and preterm infants have poor glycogen stores, whereas the infants of a diabetic mother have sufficient glycogen stores, but hyperplasia of the islet cells in the pancreas causes high insulin levels. Symptoms are jitteriness, irritability, apnoea, lethargy, drowsiness and seizures.

There is no agreed definition of hypoglycaemia in the newborn. Many babies tolerate low blood glucose levels in the first few days of life, as they are able to utilise lactate and ketones as energy stores. Recent evidence suggests that blood glucose levels above 2.6 mmol/L are desirable for optimal neurodevelopmental outcome, although during the first 24 h after birth many asymptomatic infants transiently have blood glucose levels below this level. There is good evidence that prolonged, symptomatic hypoglycaemia can cause permanent neurological disability.

Hypoglycaemia can usually be prevented by early and frequent milk feeding. In infants at increased risk of hypoglycaemia, blood glucose is regularly monitored at the bedside. If an asymptomatic infant has two low glucose values (i.e. below 2.6 mmol/L) in spite of adequate feeding or one very low value (<1.6 mmol/L) or becomes symptomatic, glucose is given by intravenous infusion aiming to maintain the glucose >2.6 mmol/L. The concentration of the intravenous dextrose may need to be increased from 10% to 15% or even 20%. Abnormal blood glucose results should be confirmed in the laboratory. High-concentration intravenous infusions of glucose should be given via a central venous catheter to avoid extravasation into the tissues, which may cause skin necrosis and reactive hypoglycaemia. If there is difficulty or delay in starting the infusion, or a satisfactory response is not achieved, glucagon or hydrocortisone can be given.

Neonatal seizures

Many babies startle or have tremors when stimulated or make strange jerks during active sleep. Seizures, on the other hand, are unstimulated. Typically, there are repetitive, rhythmic (clonic) movements of the limbs which persist despite restraint and are often accompanied by eye movements and changes in respiration. Many neonatal units now use continuous single channel EEG (amplified integrated EEG, also called a cerebral function monitor) to be able to confirm changes in electrical discharges in the brain. The causes of seizures are listed in Box 10.3.

Whenever seizures are observed, hypoglycaemia and meningitis need to be excluded or treated urgently. A cerebral ultrasound is performed to identify haemorrhage or cerebral malformation. Identification of some cerebral ischaemic lesions or cerebral malformations will require MRI scans of the brain. Treatment is directed at the cause, whenever possible. Ongoing or repeated seizures are treated with an anticonvulsant, although their efficacy in suppressing seizures is much poorer than in older children. The prognosis depends on the underlying cause.

Box 10.3 Causes of neonatal seizures

- Hypoxic-ischaemic encephalopathy
- Cerebral infarction
- Septicaemia/meningitis
- Metabolic
  - Hypoglycaemia
  - Hypo-/hypernatraemia
  - Hypocalcaemia
  - Hypomagnesaemia
  - Inborn errors of metabolism
  - Pyridoxine dependency
- Intracranial haemorrhage
- Cerebral malformations
- Drug withdrawal, e.g. maternal opiates
- Congenital infection
- Kernicterus

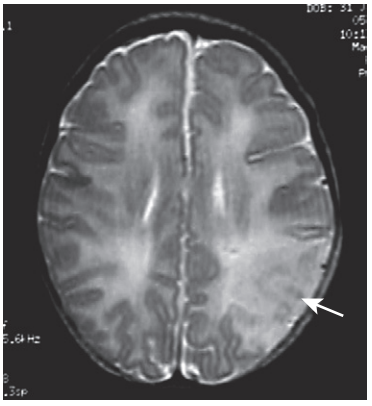


Figure 10.23 MRI scan showing infarction in the territory of a branch of the left middle cerebral artery.

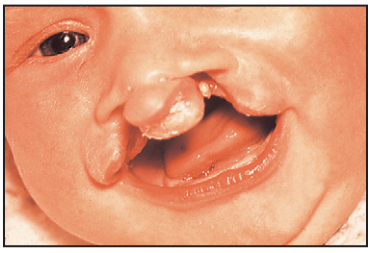
Cerebral infarction (neonatal stroke)

Infarction in the territory of the middle cerebral artery may present with seizures at 12–48 h in a term infant. The seizures may be focal or generalised. In contrast to infants with hypoxic-ischaemic encephalopathy, there are no other abnormal clinical features. The diagnosis is confirmed by MRI (Fig. 10.23). The mechanism is thought to be thrombotic, either thromboembolism from placental vessels or sometimes secondary to inherited thrombophilia. In spite of pronounced abnormalities on the MRI scans, the prognosis is relatively good, with only 20% having hemiparesis or epilepsy presenting later in infancy or early childhood.

Craniofacial disorders

Cleft lip and palate

A cleft lip (Fig. 10.24a) may be unilateral or bilateral. It results from failure of fusion of the frontonasal and maxillary processes. In bilateral cases the premaxilla is



(a)



(b)

Figure 10.24 Before **(a)** and after **(b)** operation for cleft lip. Photographs showing the impressive results of surgery help many patients cope with the initial distress at having an affected infant. (Courtesy of Mr N. Waterhouse.)

anteverted. Cleft palate results from failure of fusion of the palatine processes and the nasal septum. Cleft lip and palate affect about 0.8 per 1000 babies. Most are inherited polygenically, but they may be part of a syndrome of multiple abnormalities, e.g. chromosomal disorders. Some are associated with maternal anticonvulsant therapy. They may be detected on antenatal ultrasound scanning.

Surgical repair of the lip (Fig. 10.24b) may be performed within the first week of life for cosmetic reasons, although some surgeons feel that better results are obtained if surgery is delayed. The palate is usually repaired at several months of age. A cleft palate may make feeding more difficult, but some affected infants can still be breast-fed successfully. In bottle-fed babies, if milk is observed to enter the nose and cause coughing and choking, special teats and feeding devices may be helpful. Orthodontic advice and a dental prosthesis may help with feeding. Secretory otitis media is relatively common and should be sought on follow-up. Infants are also prone to acute otitis media. Adenoidectomy is best avoided, as the resultant gap between the abnormal palate and nasopharynx will exacerbate feeding problems and the nasal quality of speech. A multidisciplinary team approach is required, involving plastic and ENT surgeons, paediatrician, orthodontist, audiologist and speech therapist. Parent support groups can provide valuable support and advice for families (Cleft Lip and Palate Association, CLAPA).

Pierre Robin sequence

The Pierre Robin sequence is an association of micrognathia (Fig. 10.25), posterior displacement of the tongue (glossoptosis) and midline cleft of the soft palate. There may be difficulty feeding and, as the tongue falls back, there is obstruction to the upper airways which may result in cyanotic episodes. The infant is at risk of failure to thrive during the first few months. If there is upper airways obstruction, the infant may need to lie prone, allowing the tongue and small mandible to fall forward. Persistent obstruction can be treated using a nasopharyngeal airway. Eventually the mandible grows and these problems resolve. The cleft palate can then be repaired.



Figure 10.25 Micrognathia in Pierre Robin sequence.

Gastrointestinal disorders

Oesophageal atresia

Oesophageal atresia is usually associated with a tracheo-oesophageal fistula (Fig. 10.26). It occurs in 1 in 3500 live births and is associated with polyhydramnios during pregnancy. If suspected, a wide-calibre feeding tube is passed and checked by X-ray to see if it reaches the stomach. If not diagnosed at birth, clinical presentation is with persistent salivation and drooling from the mouth after birth. If the diagnosis is not made at this stage, the infant will cough and choke when fed, and have cyanotic episodes. There may be aspiration into the lungs of saliva (or milk) from the upper airways and acid secretions from the stomach. Almost half of the babies have other congenital malformations, e.g. as part of the VACTERL association (**V**ertebral, **A**norectal, **C**ardiac, **T**racheo-o**E**sophageal, **R**enal and Radial **L**imb anomalies). Continuous suction is applied to a tube passed into the oesophageal pouch to reduce aspiration of saliva and secretions pending transfer to a neonatal surgical unit.

Small bowel obstruction

This may be recognised antenatally on ultrasound scanning. Otherwise, small bowel obstruction presents with persistent vomiting, which is bile-stained unless the obstruction is above the ampulla of Vater.

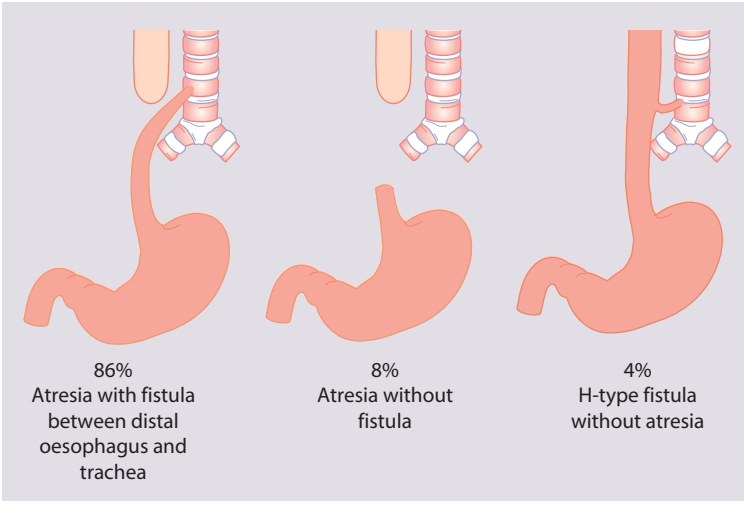


Figure 10.26 Oesophageal atresia and tracheo-oesophageal fistula.

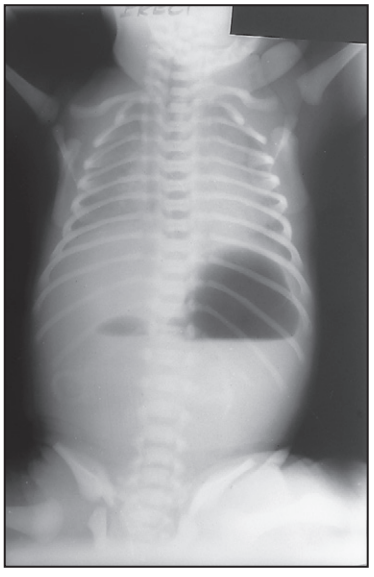


Figure 10.27 Abdominal X-ray in duodenal atresia showing a 'double bubble' from distension of the stomach and duodenal cap. There is absence of air distally.

Meconium may initially be passed, but subsequently its passage is usually delayed or absent. Abdominal distension becomes increasingly prominent the more distal the bowel obstruction. High lesions will present soon after birth, but lower obstruction may not present for some days.

Small bowel obstruction may be caused by:

- Atresia or stenosis of the duodenum (Fig. 10.27) – one-third have Down syndrome and it is also associated with other congenital malformations
- Atresia or stenosis of the jejunum or ileum – there may be multiple atretic segments of bowel
- Malrotation with volvulus – a dangerous condition as it may lead to infarction of the entire midgut
- Meconium ileus – thick inspissated meconium, of putty-like consistency, becomes impacted in the lower ileum; almost all affected neonates have cystic fibrosis
- Meconium plug – a plug of inspissated meconium causes lower intestinal obstruction.

The diagnosis is made on clinical features and abdominal X-ray showing intestinal obstruction. Atresia or stenosis of the bowel and malrotation are treated surgically, after correction of fluid and electrolyte depletion. A meconium plug will usually pass spontaneously. Meconium ileus may be dislodged using Gastrografin contrast medium.

Large bowel obstruction

This may be caused by:

- *Hirschsprung disease*. Absence of the myenteric nerve plexus in the rectum which may extend along the colon. The baby often does not pass meconium within 48 h of birth and subsequently the abdomen distends. About 15% present as an acute enterocolitis (see Ch. 13).
- *Rectal atresia*. Absence of the anus at the normal site. Lesions are high or low, depending whether the bowel ends above or below the levator ani muscle. In high lesions, there is a fistula to the bladder or urethra in boys, or adjacent to the vagina or to the bladder in girls. Treatment is surgical.

Bile-stained vomiting is from intestinal obstruction until proved otherwise.

Exomphalos/gastroschisis

These lesions are often diagnosed antenatally (see Ch. 9). In exomphalos (also called omphalocele), the abdominal contents protrude through the umbilical ring, covered with a transparent sac formed by the amniotic membrane and peritoneum (Fig. 10.28). It is often associated with other major congenital abnormalities. In gastroschisis, the bowel protrudes through a defect in the anterior abdominal wall, adjacent to



Figure 10.28 Small exomphalos with loops of bowel confined to the umbilicus. Care needs to be taken not to put a cord clamp across these lesions.

the umbilicus, and there is no covering sac (see Fig. 9.2). It is not associated with other congenital abnormalities.

Gastroschisis carries a much greater risk of dehydration and protein loss, so the abdomen of affected infants should be wrapped in several layers of clingfilm to minimise fluid and heat loss. A nasogastric tube is passed and aspirated frequently and an intravenous infusion of dextrose established. Colloid support is often required to replace protein loss. Many lesions can be repaired by primary closure of the abdomen. With large lesions, the intestine is enclosed in a silastic sac sutured to the edges of the abdominal wall and the contents gradually returned into the peritoneal cavity.

Further reading

Lissauer T, Fanaroff AA: *Neonatology at a Glance*, ed 2, Oxford, 2011, Blackwell Science.

Short, illustrated textbook.

Rennie JM: *Roberton's Textbook of Neonatology*, ed 4, Edinburgh, 2011, Elsevier/Churchill Livingstone.

Comprehensive textbook.

Websites (Accessed May 2011)

BLISS. For parents of infants born too early, too small or too sick: Available at: http://www.bliss.org.uk/.

Sands. Stillbith and neonatal death charity: Available at: http://www.uk-sands.org/.

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11

Growth and puberty

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In contrast to adults, growth of all body parameters and, later, the development of puberty are key features of childhood and adolescence. Deviation from the norm needs to be recognised and the underlying cause identified and treated. This requires knowledge concerning normal growth and puberty.

There are four phases of normal human growth (Fig. 11.1).

Fetal

This is the fastest period of growth, accounting for about 30% of eventual height. Size at birth is determined by the size of the mother and by placental nutrient supply, which in turn modulates fetal growth factors (IGF-2, human placental lactogen and insulin). Optimal placental nutrient supply is dependent on an adequate maternal diet. Size at birth is largely independent of the father's height and of growth hormone. Severe intrauterine growth restriction and extreme prematurity when accompanied by poor postnatal growth can result in permanent short stature. Paradoxically, low birthweight increases the later metabolic risk of childhood obesity.

The infantile phase

Growth during infancy to around 18 months of age is also largely dependent on adequate nutrition. Good health and normal thyroid function are also necessary. This phase is characterised by a rapid but decelerating growth rate, and accounts for about 15% of eventual height. By the end of this phase, children have changed from their fetal length, largely determined by the uterine environment, to their genetically determined height. An inadequate rate of weight gain during this period is called 'failure to thrive' (see Ch. 12).

Childhood phase

This is a slow, steady but prolonged period of growth that contributes 40% of final height. Pituitary growth hormone (GH) secretion acting to produce insulin-like growth factor 1 (IGF-1) at the epiphyses is the main determinant of a child's rate of growth, provided there is adequate nutrition and good health. Thyroid hormone, vitamin D and steroids also affect cartilage cell division and bone formation. Profound chronic unhappiness can decrease GH secretion and accounts for psychosocial short stature.

Pubertal growth spurt

Sex hormones, mainly testosterone and oestradiol, cause the back to lengthen and boost GH secretion. This adds 15% to final height. The same sex steroids cause fusion of the epiphyseal growth plates and a cessation of growth. If puberty is early, which is not uncommon in girls, the final height is reduced because of early fusion of the epiphyses (see below).

Measurement

Growth must be measured accurately, with attention to correct technique and accurate plotting of the data:

- Weight – readily and accurately determined with electronic scales but must be performed on a naked infant or a child dressed only in underclothing as an entire month's or year's weight gain can be represented by a wet nappy or heavy jeans, respectively.
- Height – the equipment must be regularly calibrated and maintained. In children over 2 years

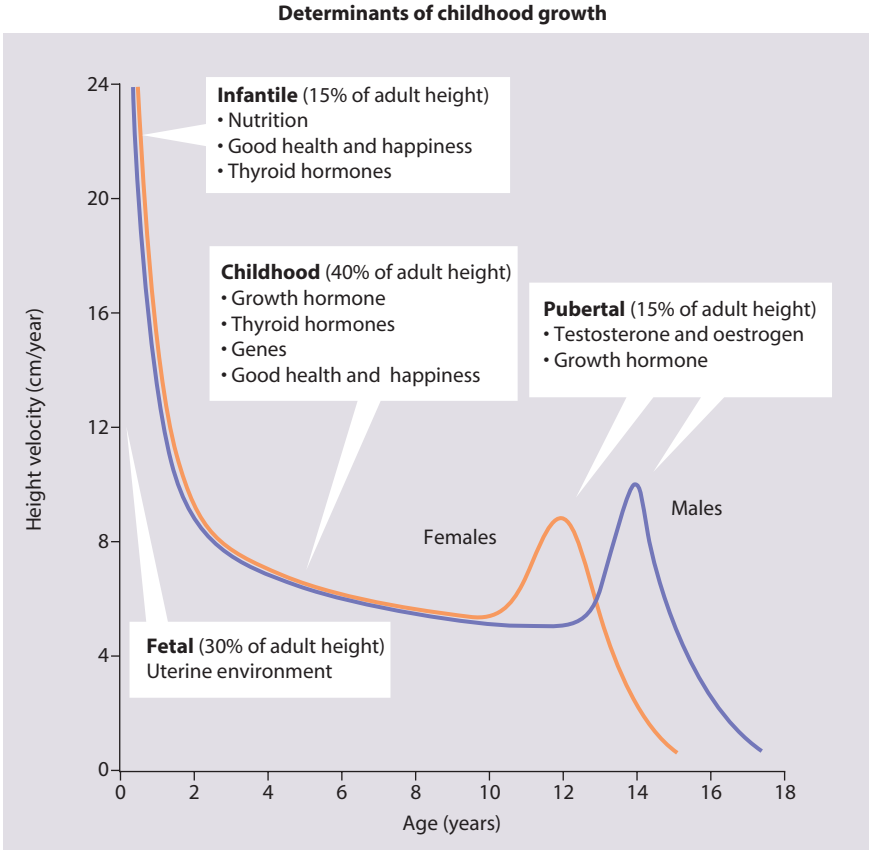


Figure 11.1 Male and female height velocity charts (50th percentile) showing the determinants of childhood growth. The fetal and infantile phases are mainly dependent on adequate nutrition, whereas the childhood and pubertal phases are dependent on growth hormone and other hormones. Adult males are taller than females as they have a longer childhood growth phase, their peak height velocity is higher and their growth ceases later.

of age, the standing height is measured as illustrated in Figure 11.2. In children under 2 years, length is measured lying horizontally (Fig. 11.3), using the mother to assist. Accurate length measurement in infants can be difficult to obtain, as the legs need to be held straight and infants often dislike being held still. For this reason, routine measurement of length in infancy is often omitted from child surveillance, but it should always be performed whenever there is doubt about an infant's growth.

- Head circumference – the occipitofrontal circumference is a measure of head and hence brain growth. The maximum of three measurements is used. It is of particular importance in developmental delay or suspected hydrocephalus

These measurements should be plotted as a simple dot on an appropriate growth centile chart. Standards for a population should be constructed and updated every generation to allow for the trend towards earlier puberty and taller adult stature from improved childhood nutrition. In 2009, the UK adopted the World Health Organization (WHO) new global Child Growth Standards for infants and children 0–4 years old (See Appendix Fig. A1). The new charts are based on the optimal growth of healthy children totally

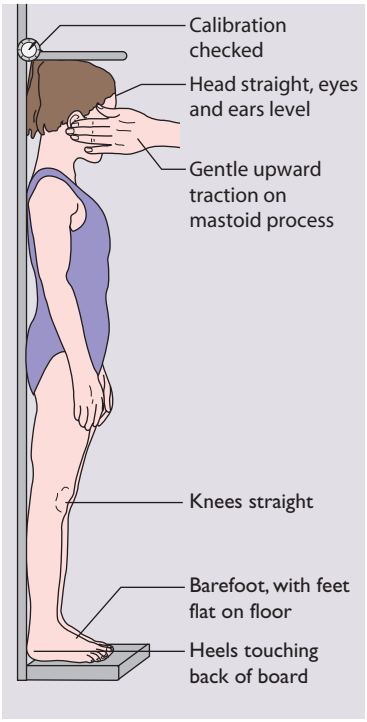


Figure 11.2 Measuring height accurately in children.

breast-fed up to the age of 6 months. These charts allow for the lower weight of totally breast-fed infants and are therefore less likely to identify some breast-fed babies as underweight and may also allow early identification of bottle-fed babies gaining weight too rapidly.

Height in a population is normally distributed and the deviation from the mean can be measured as a centile or standard deviation (Fig. 11.4). The bands on the growth reference charts have been chosen to be two-thirds of a standard deviation apart and correspond approximately to the 25th, 9th, 2nd and 0.4th centiles below the mean, and the 75th, 91st, 98th and 99.6th centiles above the mean. The further these centiles lie from the mean, the more likely it is that a child has a pathological cause for his short or tall stature. For instance, values below the 0.4th or above the 99.6th centile will occur by chance in only 4 per 1000 children and can be used as a criterion for referral from primary to specialist care. A single growth parameter should not be assessed in isolation from the other growth parameters: e.g. a child's low weight may be in proportion to the height if short, but abnormal if tall. Serial measurements are used to show the pattern and determine the rate of growth. This is helpful in diagnosing or monitoring many paediatric conditions. The WHO charts include an adult height predictor and a BMI centile ready-reckoner.

Summary

Measurement of children

- Measurement must be accurate for meaningful monitoring of growth
- Growth parameters should be plotted on charts
- Significant abnormalities of height are:
  - measurements outside the 0.4th or 99.6th centiles if the mid-parental height is not short or tall
  - if markedly discrepant from weight
  - serial measurements which cross growth centile lines after the first year of life.

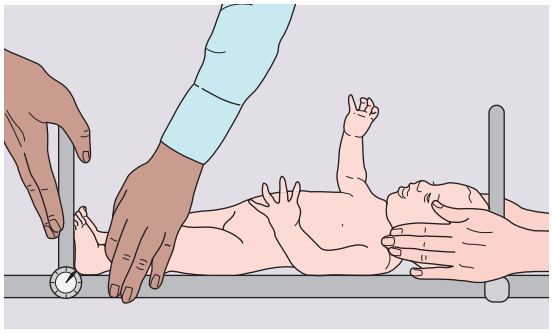


Figure 11.3 Measuring length in infants and young children. An assistant is required to hold the legs straight.

Puberty

Puberty follows a well-defined sequence of changes that may be assigned stages, as shown in Figures 11.5 and 11.6. Over the last 20 years, the mean age at which puberty starts in girls has lowered. However, the age at which menarche occurs has remained stable. Therefore, females now remain in puberty for longer.

In *females* the features of puberty are:

- Breast development – the first sign, usually starting between 8.5 and 12.5 years
- Pubic hair growth and a rapid height spurt – occur almost immediately after breast development
- Menarche – occurs on average 2.5 years after the start of puberty and signals that growth is coming to an end, with only around 5 cm height gain remaining.

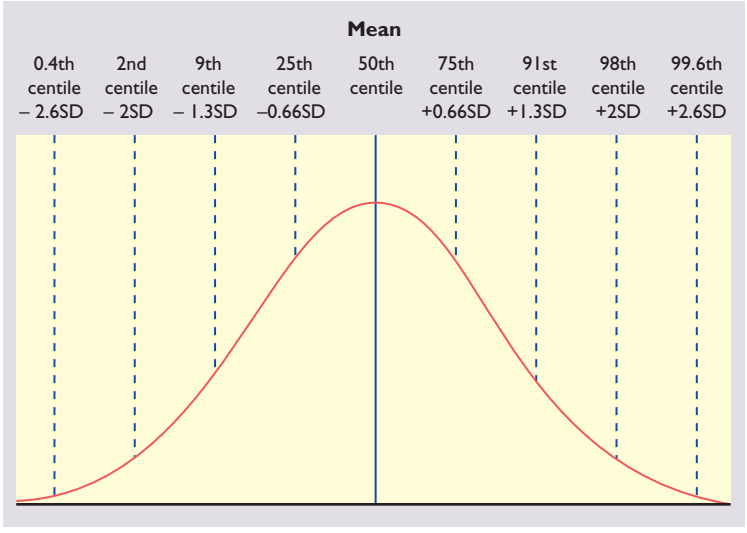


Figure 11.4 Interpretation of the UK growth reference charts. The lines show the mean and bands which are two-thirds of a standard deviation (SD) apart. The centiles are shown in the diagram.

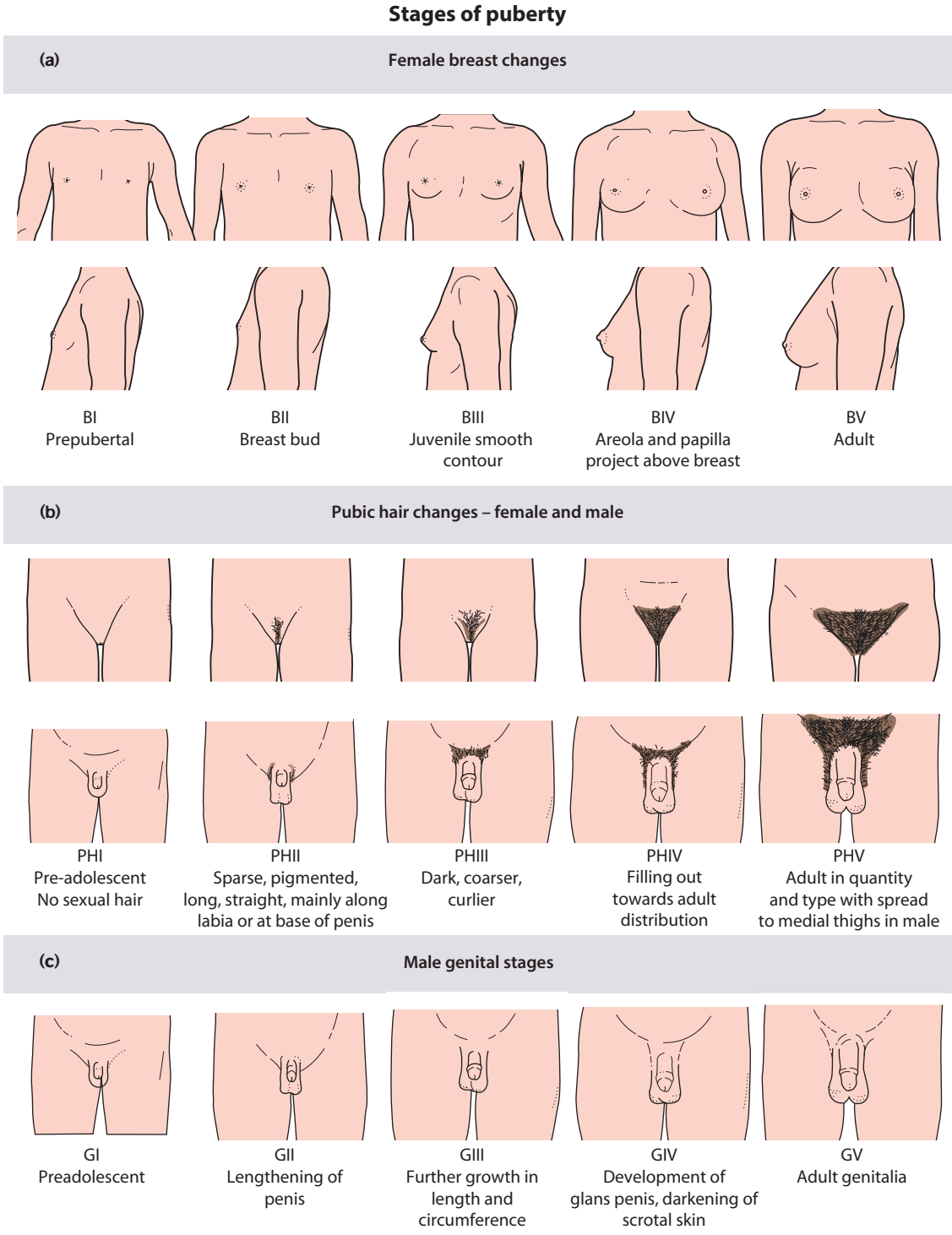


Figure 11.5 Schematic drawings of male and female stages of puberty. Pubertal changes are shown according to the Tanner stages of puberty.

In *males*:

- Testicular enlargement to >4 ml volume measured using an orchidometer (Fig. 11.7) – the first sign of puberty
- Pubic hair growth – follows testicular enlargement, usually between 10 and 14 years of age
- Height spurt – when the testicular volume is 12–15 ml, after a delay of around 18 months.

The height spurt in males occurs later and is of greater magnitude than in females, accounting for the greater final average height of males than females.

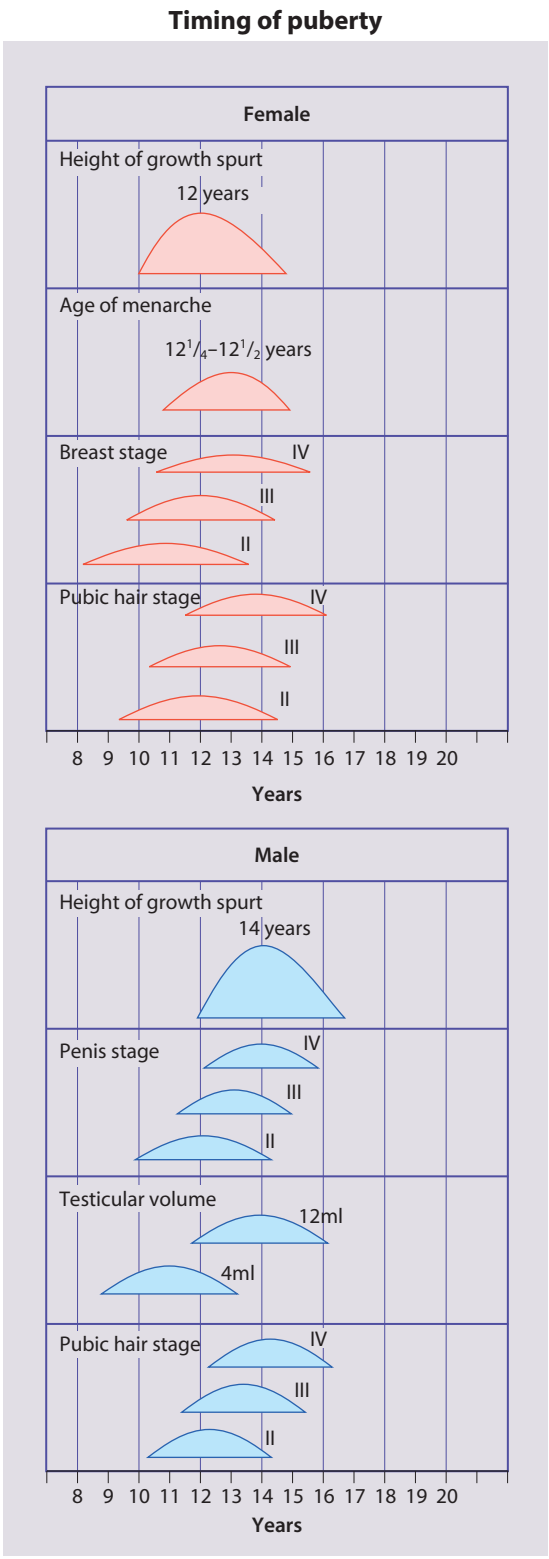


Figure 11.6 Timing of puberty. Pubertal changes are shown according to the Tanner stages of puberty. (Diagrams based on Zitelli BJ, Davis HW. 1992. *Atlas of Pediatric Physical Diagnosis,* 2nd edn. Lippincott, Philadelphia; Johnson TR, Moore WM, Jeffries JE. 1978. *Children are Different,* 2nd edn. Ross Laboratories, Division of Abbot Laboratories, Columbus, OH.)



Figure 11.7 Orchidometer to assess testicular volume (in ml). (From Wales JKH, Rogol AD, Wit JM. 2003. *Pediatric Endocrinology and Growth*. Mosby-Wolfe, London, with permission.)

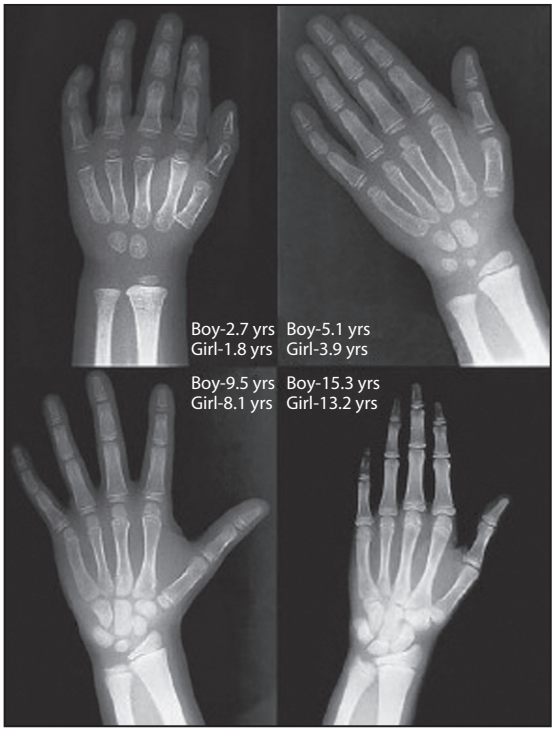


Figure 11.8 X-rays of the left wrist and hand to determine bone ages. This technique allows assessment of skeletal maturation from the time of appearance or maturity of the epiphyseal centres, using a standardised rating system. The child's height can be compared with skeletal maturation and an adult height prediction made. The ages show bone age of each X-ray.

In *both sexes*, there will be development of acne, axillary hair, body odour and mood changes.

If puberty is abnormally early or late, it can be further assessed:

- Bone age measurement from a hand and wrist X-ray to determine skeletal maturation (Fig. 11.8)
- In females, pelvic ultrasound can be used to assess uterine size and endometrial thickness.